EE 3072
System Dynamics and Control

Lecture 3

Time-domain analysis of control systems

OUTLINE

Why worry about time-domain response of a control system?

Time response of first-order systems
- Properties of first-order systems

Time response of a prototype 2nd-order system
- Properties of 2nd-order systems
- The effect of pole locations on 2nd-order systems
- Effect of adding poles and zeros

OUTLINE

Stability of linear systems
(in complex plane)

Routh-Hurwithz Criterion

Steady-state error

Why worry about time-domain response?

What is Time Response Behavior?

- Since time is used as an independent variable in most control systems, it is usually of interest to evaluate the output response with respect to time, or simply, the time response.

- When you design a system, the time response behavior may well be the most important aspect of its' behavior.

Points you might worry about include

- How quickly a system responds is important.
  - If you have a control system that's controlling a temperature, how long it takes the temperature to reach a new steady state is important.
- Overshoot and how close a system comes to instability.
  - Say you're trying to control a temperature, and you want the temperature to be 200° C. If the temperature goes to 250° C before it settles out, you'll want to know that.
- Oscillations in a system are not usually desirable
  - If you're trying to control speed of an automobile at 55kph and the speed keeps varying between 50kph and 60kph, your design isn't very good.
- Deviations from the desired value in a system are not usually desirable
  - If you're trying to control speed of an automobile at 55kph and the speed settles at 50kph, your design isn't very good.

Points you might worry about include

These are a few of many different aspects of time behavior of a system that are important in control system design.

The examples above really are talking about aspects like:

- Speed of response
- Relative stability of the system
- Stability of the system
- Steady state error

In this lecture…

- We will begin to examine how it is possible to predict aspects of the time behavior of a system.

- We'll do that by starting with a first order system and examining the parameters of that kind of system that control their time behavior.

Time response of first-order systems

- What is a first-order system?
  - First order systems are described by first order differential equations.

Example

First-order differential equation: $\tau \frac{dy(t)}{dt} + y(t) = Ku(t)$

y(t)---output response of the system; u(t)---input to the system

Using Laplace transform and assuming zero initial conditions, we get:

$$\tau s Y(s) + Y(s) = KU(s)$$

Transfer function: $G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}$

τ - time constant

K - DC gain

Why learn about first order systems?

- First-order systems are the simplest systems, and they make a good place to begin a study of system dynamics.

- First-order system concepts form the foundation for understanding more complex systems.

- Everything starts here…

A number of goals

- First, if you have a first-order system, you need to be able to predict and understand how it responds to an input, so you need to be able to do this.

Given a first-order system



Determine the impulse and step response of the system.

A number of goals

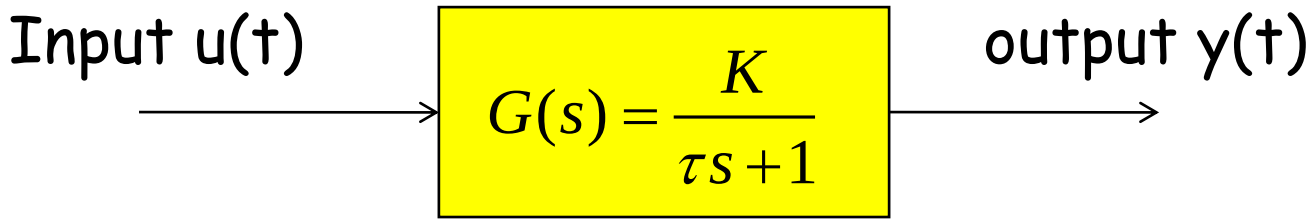
- Second, you may go into a lab and measure a system, and if it is first order, you need to be able to do this.

Given the time response of a first order system



Determine the parameters of the system (time constant and DC gain)

Parameters of a 1-order system

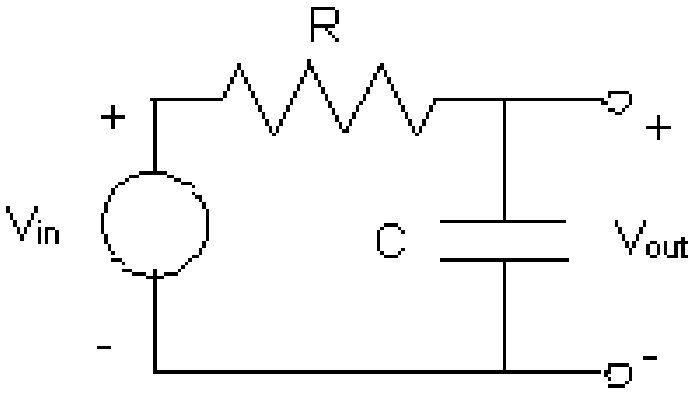


τ, the time constant, will determine how quickly the system moves toward steady state.

K, the DC gain of the system, will determine the size of steady state response when the output settles out to a constant value.

Some example systems

A Resistor-Capacitor Circuit



Differential equation:

$$RC\frac{dV_{out}}{dt} + V_{out} = V_{in}$$

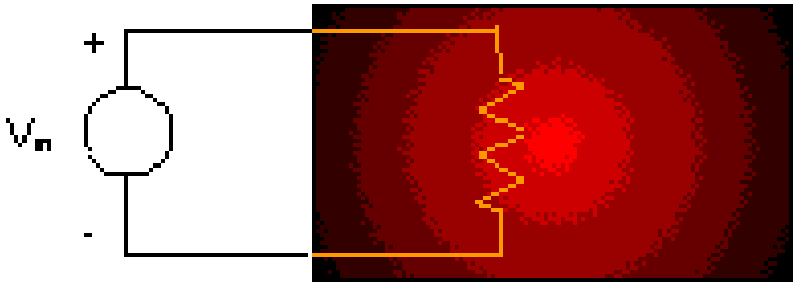
Transfer function:

$$G(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{RCs+1}$$

Some example systems

A Simple Thermal System

In this system, heat flows into a heated space and the temperature within the heated space follows a first order linear differential equation.

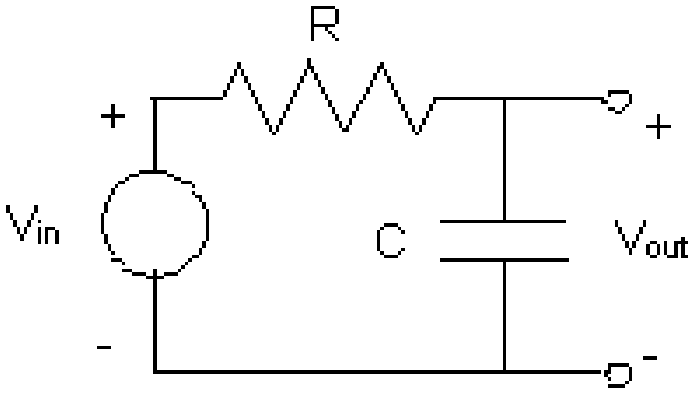


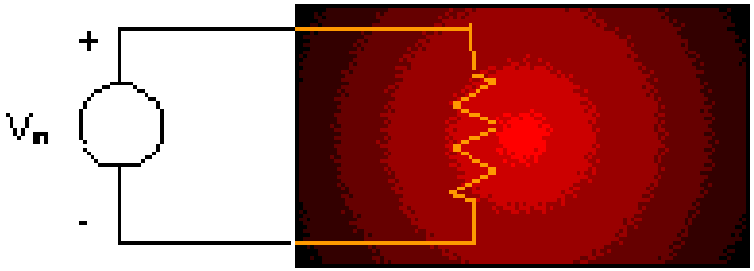
Differential equation:

$$\tau \frac{dTemp}{dt} = -Temp + K \cdot Heat_{in}$$

Transfer function:

$$G(s) = \frac{Temp(s)}{Heat_{in}(s)} = \frac{K}{\tau s + 1}$$





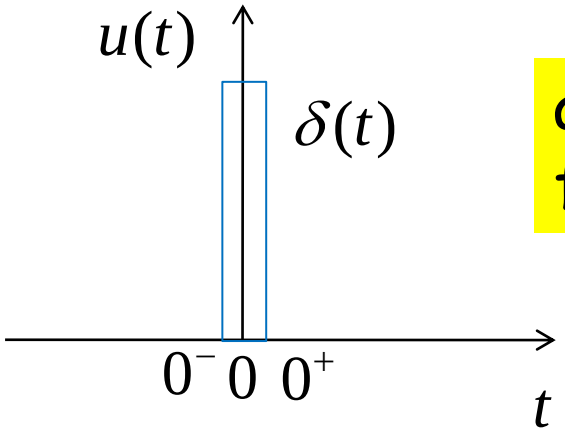
- The systems above come from very diverse places, including circuit theory, thermal dynamics, etc.
- However, there is a common mathematical description for all those systems.
- That's why you need to learn the properties of a general first-order system.

- When you learn about first-order system dynamics you are learning a topic that:
  - Has applicability to a wide variety of areas
  - Is a good introduction to more complex system dynamics, like second-order systems and more complex systems of higher order.

- We'll start by learning how a first order system responds to two inputs:
  - unit impulse input
  - unit step input

Impulse Response of a 1-Order System

- The impulse response is the response to a unit impulse input $\delta(t)$.



a very intense force for a very short time

- The unit impulse has a Laplace transform of unity (1).

$$L[\delta(t)] = 1$$

- That gives the unit impulse a unique stature.

First-order systems satisfy this generic differential equation

$$\tau \frac{dy(t)}{dt} + y(t) = Ku(t)$$

τ - time constant

K - DC gain

For a unit impulse input $\delta(t)$ and assuming zero initial conditions, could you calculate its output?

$$y(t) = \frac{K}{\tau} e^{-t/\tau}$$

The impulse response is the inverse transform of the transfer function of the above system:

$$G(s) = \frac{K}{\tau s + 1}$$

Now, we need to examine what the impulse response looks like…

Example 1

Consider a first-order system with the following parameters.

Time constant $\tau = 0.1s$; DC gain $K = 20$

The problem is to determine the unit impulse response of a system that has these parameters.

Solution. **Using the general form of the impulse response of first –order systems**

$$y(t) = \frac{K}{\tau} e^{-t/\tau}$$

With the parameters above, the impulse response is:

$$y(t) = \frac{K}{\tau} e^{-t/\tau} = \frac{20}{0.1} e^{-t/0.1} = 200e^{-10t}$$

What value does the impulse response start from?

Example 1

Consider a first-order system with the following parameters.

Time constant $\tau = 0.1s$; DC gain $K = 20$

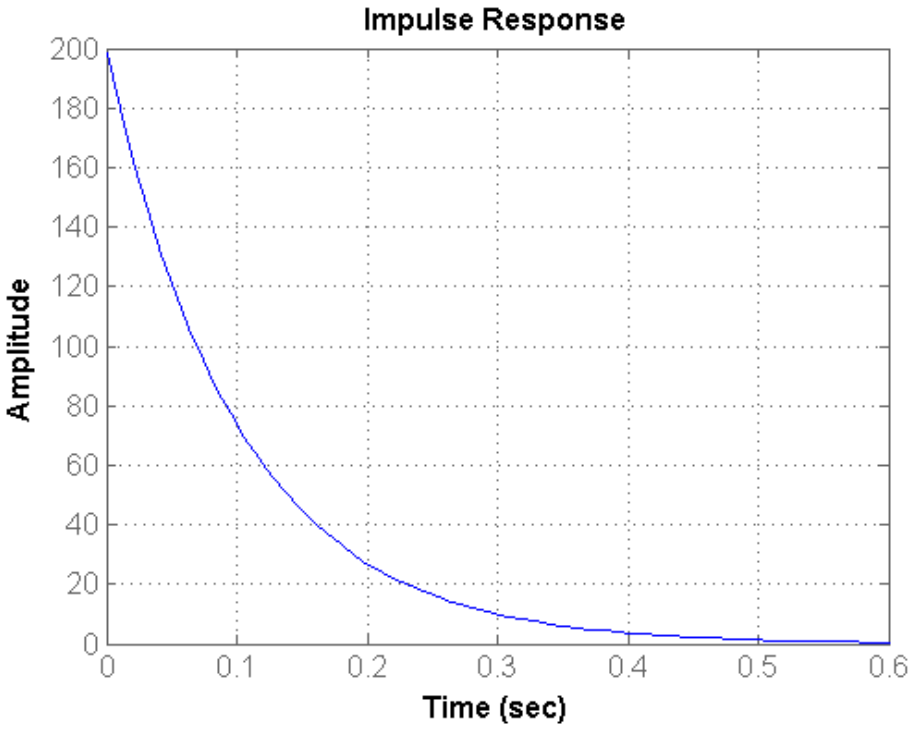
The problem is to determine the unit impulse response of a system that has these parameters.

Solution. $$y(t) = \frac{K}{\tau} e^{-t/\tau} = \frac{20}{0.1} e^{-t/0.1} = 200e^{-10t}$$

Using MATLAB to get its impulse response

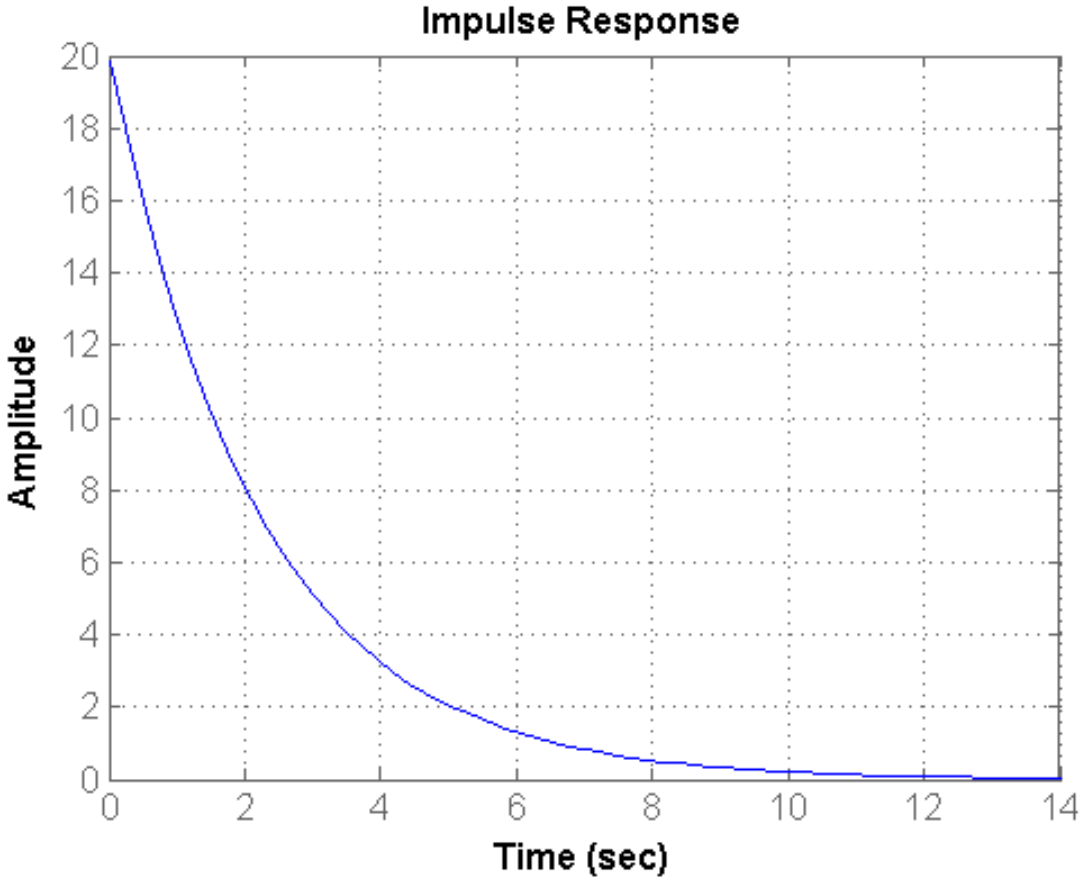
$$G(s) = \frac{K}{\tau s + 1} = \frac{20}{0.1s + 1}$$

```
>> num=20;
>> den=[0.1 1];
>> impulse(num,den)
```



Example 2

Below is the impulse response of a system - i.e. the response to a unit impulse.



The system starts with an initial condition of zero just before the impulse comes along at t = 0, so y(0-) = 0.

Could you compute the parameters of the system?

Solution.

The general time response of a first-order system is

$$y(t) = \frac{K}{\tau} e^{-t/\tau} \qquad (1)$$

From the right figure, we have

$$\text{at } t = 0,\ y(0) = 20$$

So we can get

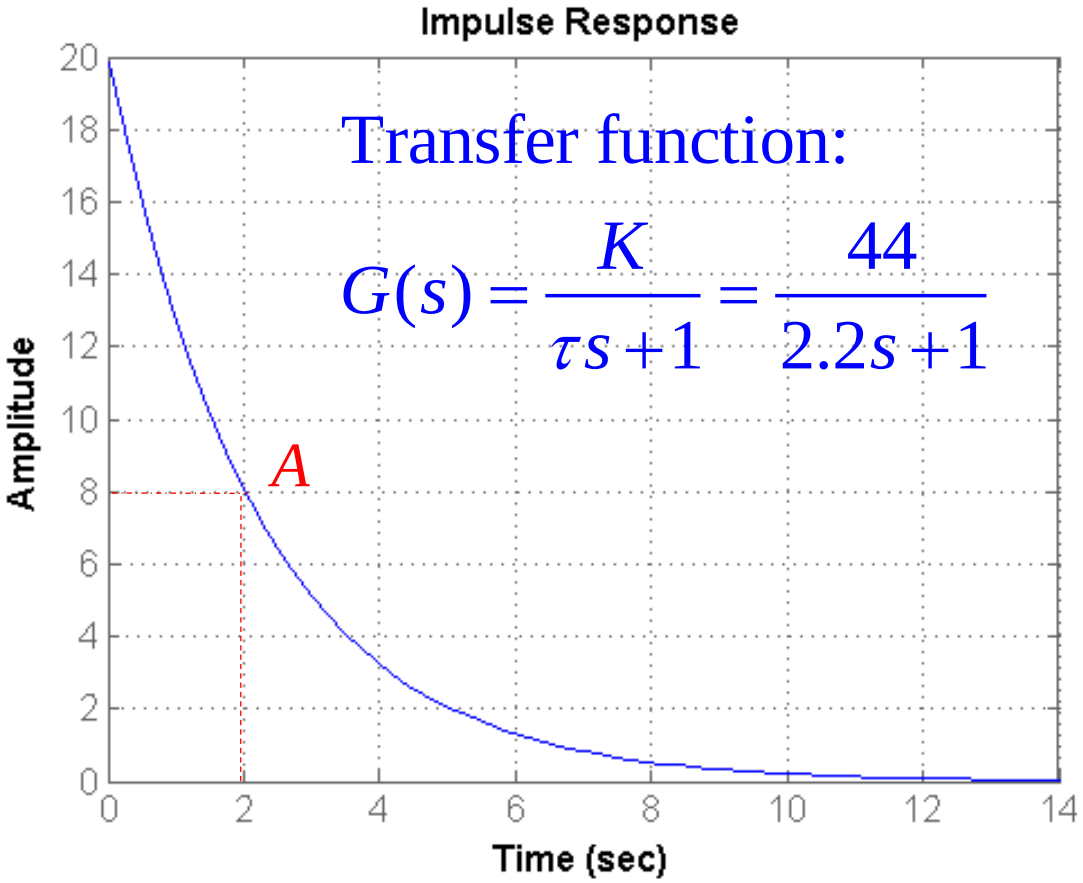
$$\frac{K}{\tau} = 20 \qquad (2)$$

How to get *K* ?

Pick up a point on the response curve:

***A*:** $\text{at } t = 2\text{sec}; y(2) = 8$

$$y(2) = \frac{K}{\tau} e^{-2/\tau} = 8 \qquad (3)$$



Transfer function:

$$G(s) = \frac{K}{\tau s + 1} = \frac{44}{2.2s + 1}$$

With (2) and (3), we have

$$y(2) = 20e^{-2/\tau} = 8$$

Solving the equation yields: $e^{-2/\tau} = 0.4$

$$-2/\tau = \ln(0.4) = -0.9163$$

$$\tau \approx 2.2\text{sec} \qquad K = 20\tau = 44$$

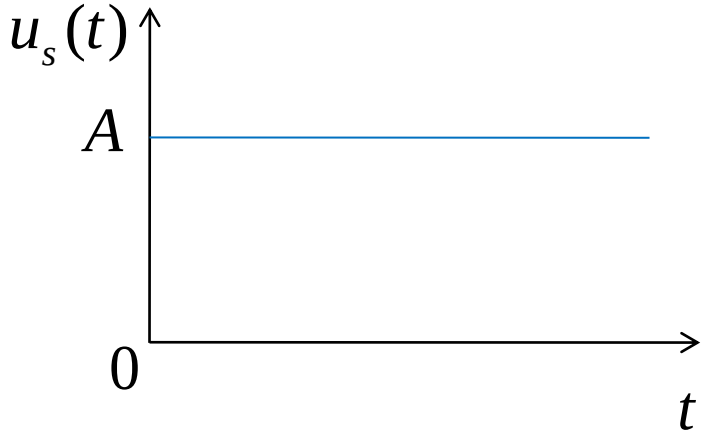
A few conclusions from examples

- Calculating the impulse response is straight-forward.
  - Given the system parameters it is not difficult to calculate - predict - the response of the system.
- The inverse problem is somewhat more difficult.
  - Given a response, you will have to be more inventive to determine what the system was that produced the given response - the system identification problem.
- The underlying theory is the same.
  - You use the same general principles to solve both problems, but the way you have to use the information makes the identification problem more difficult.

Step Response of a 1-Order System

- The step response is the response to a step input $u_s(t)$.

$$u_s(t) = \begin{cases} A & t \geq 0 \\ 0 & t < 0 \end{cases}$$



When the magnitude of the step input is 1, it is called a unit-step input, denoted by $1(t)$.

- The Laplace transform of the unit step $1(t)$ is

$$L[1(t)] = \frac{1}{s}$$

First-order systems satisfy this generic differential equation

$$\tau \frac{dy(t)}{dt} + y(t) = Ku(t)$$

τ - time constant

K - DC gain

For a unit step input $u(t) = 1(t)$ and assuming zero initial conditions, could you calculate its output?

$$\tau s Y(s) + Y(s) = KU(s)$$

$$U(s) = L[1(t)] = \frac{1}{s}$$

$$\Rightarrow \quad (\tau s + 1)Y(s) = \frac{K}{s}$$

$$\Rightarrow \quad Y(s) = \frac{K}{s(\tau s + 1)} = K\left(\frac{1}{s} - \frac{\tau}{\tau s + 1}\right) \quad \Rightarrow \quad y(t) = K(1 - e^{-t/\tau})$$

$t = 0 \quad y(t) = 0$

$t \to \infty \quad y(t) \to K$

Can you imagine what the response looks like?

Example 3

Consider a first-order system with the following parameters.

Time constant $\tau = 0.1s$; DC gain $K = 20$

The problem is to determine the unit step response of a system that has these parameters.

Solution.

Using the general form of the unit-step response of first –order systems

$$y(t) = K(1 - e^{-t/\tau})$$

With the parameters above, the impulse response is:

$$y(t) = 20(1 - e^{-t/0.1}) = 20(1 - e^{-10t})$$

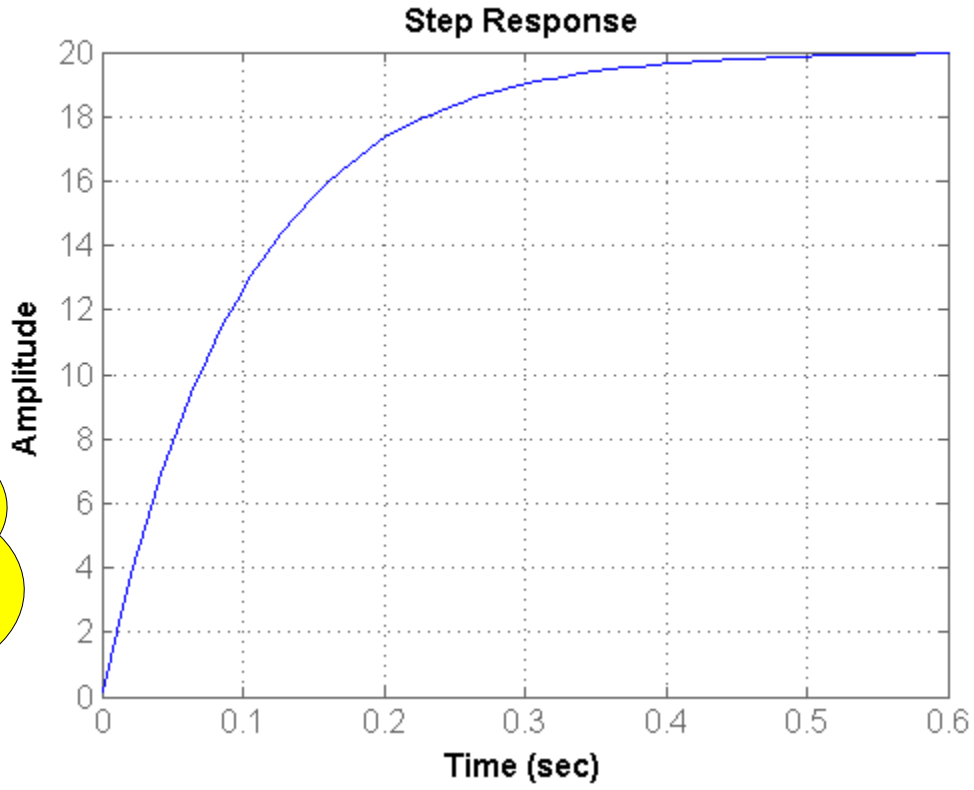
Example 3 Time constant $\tau = 0.1s$; DC gain $K = 20$

Solution. **Unit-step time response:** $y(t) = 20(1 - e^{-10t})$

Transfer function of the first-order system: $$G(s) = \frac{K}{\tau s + 1} = \frac{20}{0.1s + 1}$$

Using MATLAB to get its step response

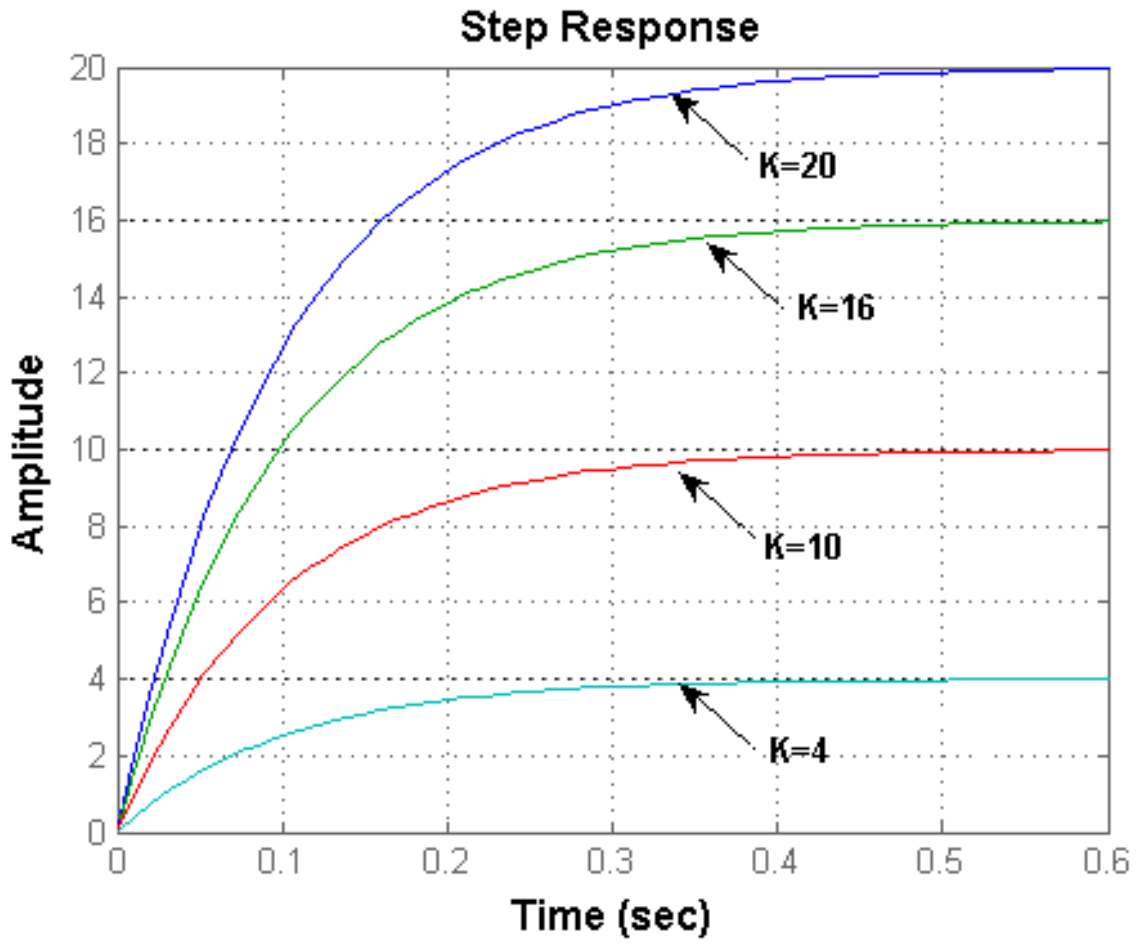
```
>> num=20;
>> den=[0.1 1];
>> step(num,den)
```



If we keep $\tau = 0.1s$ but change K, what will happen?

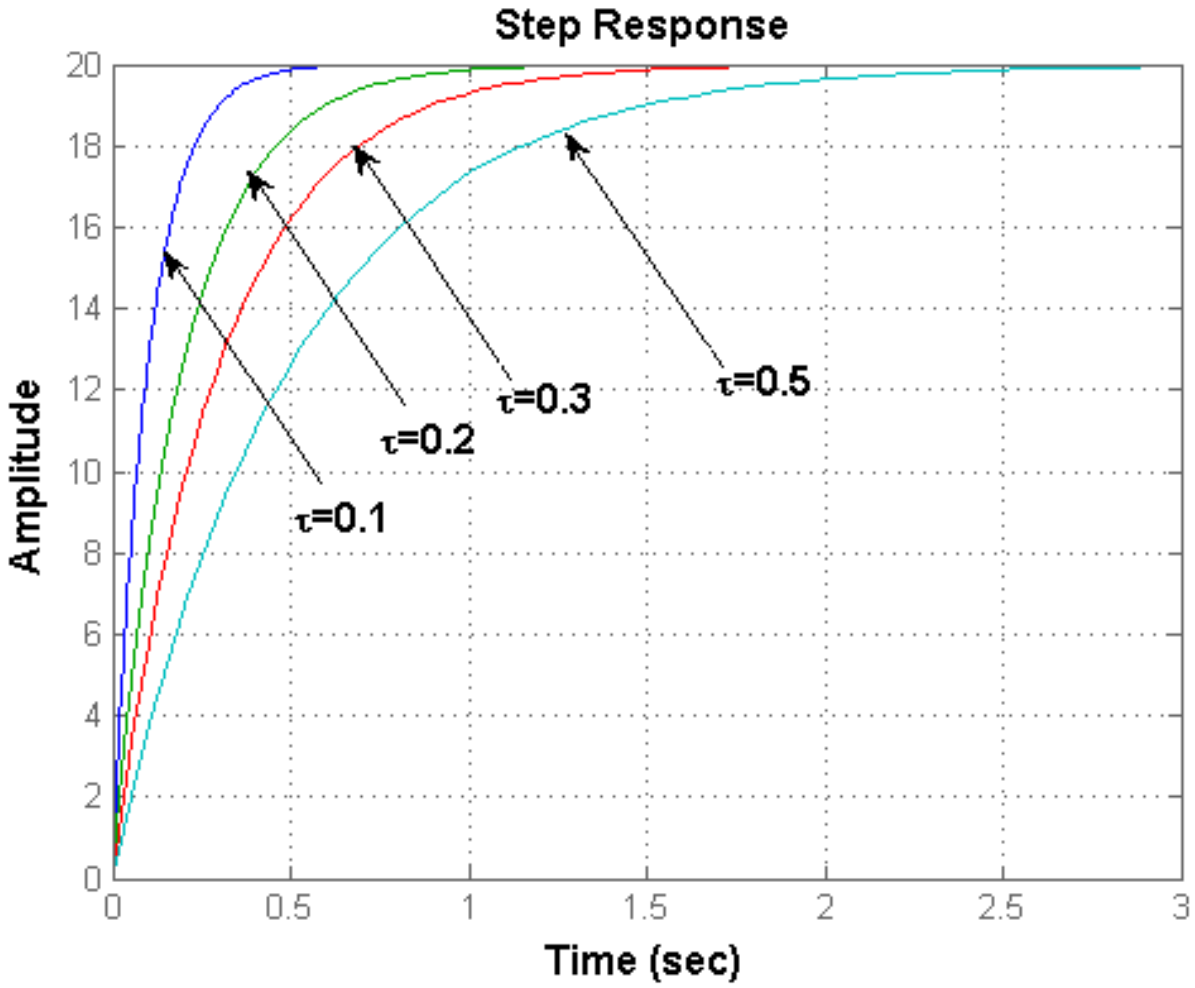
Time constant $\tau = 0.1s$;

DC gain $K = 20 \rightarrow 16 \rightarrow 10 \rightarrow 4$



DC gain $K = 20$

Time constant $\tau = 0.1s \rightarrow 0.2s \rightarrow 0.3s \rightarrow 0.5s$



Note: the time constant reflects the system inertia. The smaller is the system inertia, the shorter is τ and the quicker is the response, *vice versa*.

Two important properties of the unit-step response of a first-order system:

a) The time constant τ can be used to calculate the system output

$$y(t) = K(1 - e^{-t/\tau})$$

$t = \tau,\ y(\tau) = 0.632\,y(\infty) = 63.3\%$ final value;

$t = 2\tau,\ y(2\tau) = 0.865\,y(\infty) = 86.5\%$ final value;

$t = 3\tau,\ y(3\tau) = 0.950\,y(\infty) = 95.0\%$ final value;

$t = 4\tau,\ y(4\tau) = 0.982\,y(\infty) = 98.2\%$ final value;

$t = 5\tau,\ y(5\tau) = 0.995\,y(\infty) = 99.5\%$ final value;

experiment methods

-- estimate the time constant

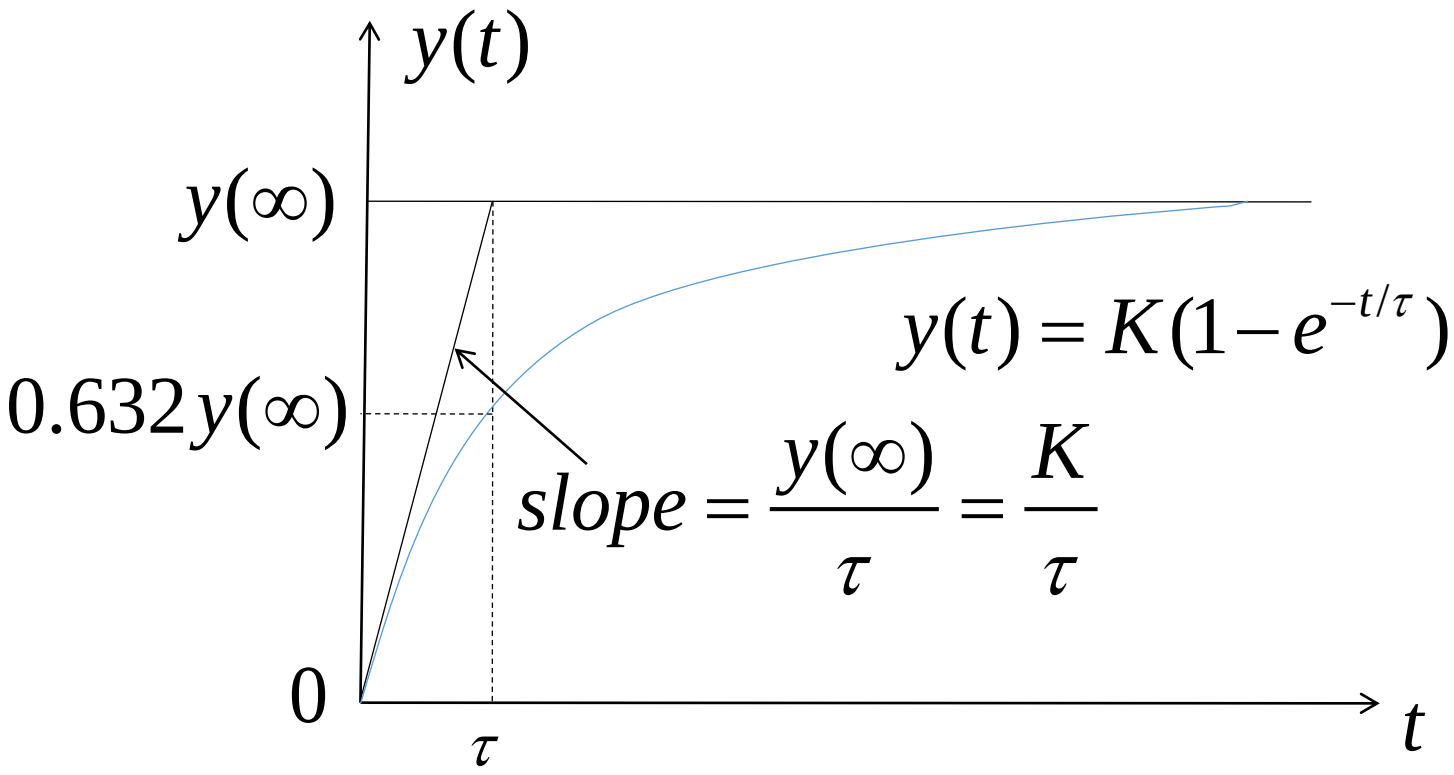
--judge whether a system is first-order or not

b) The initial slope of the response curve is K/τ and the slope decreases with time

$$\left.\frac{dy(t)}{dt}\right|_{t=0} = \frac{K}{\tau}, \quad \left.\frac{dy(t)}{dt}\right|_{t=\tau} = 0.368\frac{K}{\tau}, \quad \left.\frac{dy(t)}{dt}\right|_{t=\infty} = 0.$$

It is also a common method to obtain the time constant though the initial slope in control engineering.

Unit-step response of a first-order system

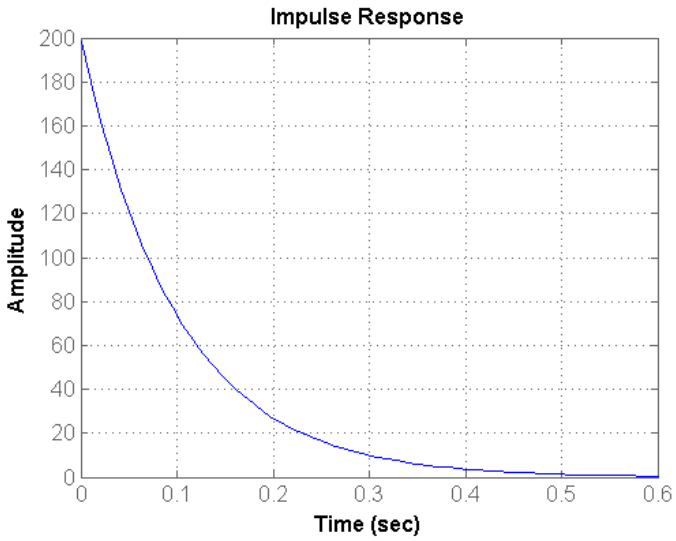


Summary: 1-order Systems

$$G(s) = \frac{K}{\tau s + 1}$$

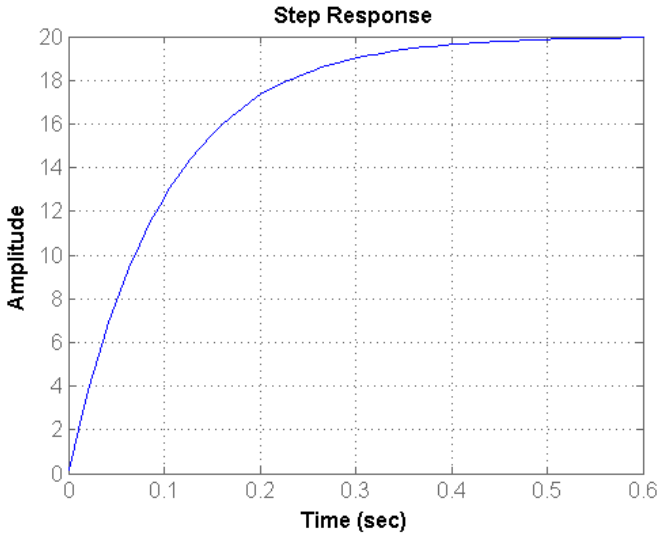
Unit-impulse response:

$$y(t) = \frac{K}{\tau} e^{-t/\tau}$$



Unit-step response:

$$y(t) = K(1 - e^{-t/\tau})$$



Summary: 1-order Systems

Unit-impulse response:

$$y(t) = \frac{K}{\tau} e^{-t/\tau}$$

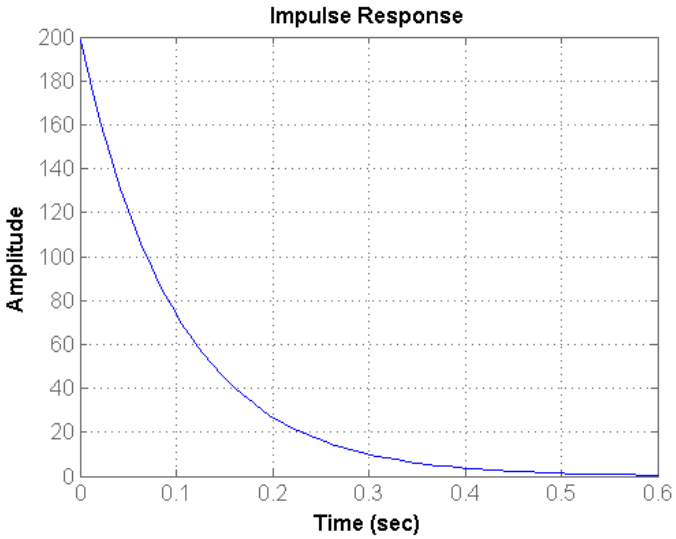
Take the Integral

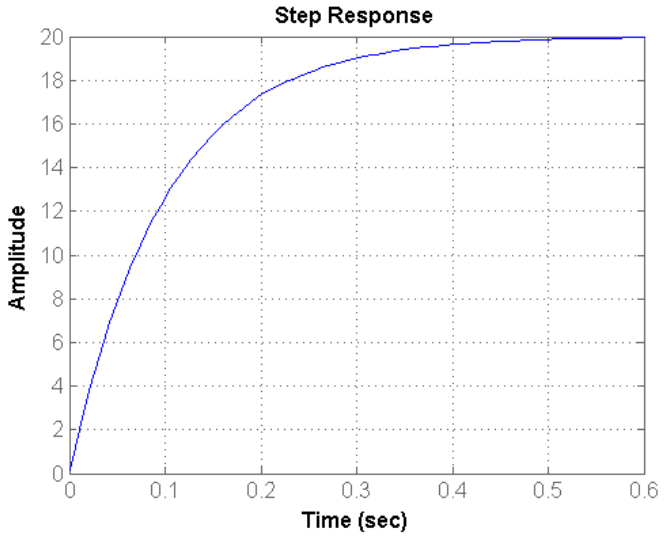
Take the Derivative

Unit-step response:

$$y(t) = K(1 - e^{-t/\tau})$$

Property of all LTI systems.





Time response of 2-order systems

- What is a second-order system?
  - Second-order systems are described by second-order differential equations.

Example

A prototype second-order differential equation:

$$\frac{d^2}{dt^2}y(t)+2\zeta\omega_n\frac{d}{dt}y(t)+\omega_n^2 y(t)=\omega_n^2 u(t)$$

$y(t)$---output response of the system;

$u(t)$---input to the system

Using Laplace transform and assuming zero initial conditions, we get:

$$s^2Y(s) + 2\zeta\omega_n sY(s) + \omega_n^2 Y(s) = \omega_n^2 U(s)$$

Transfer function of a second-order system:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ζ - damping ratio , will determine how much the system oscillates as the response decays toward steady state.

ω_n - undamped natural frequency , will determine how fast the system oscillates during any transient response

Step response of second-order systems

A 2-order system: $G(s)=\dfrac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

Input : $u(t)=1(t);\quad U(s)=\dfrac{1}{s}$

Case 1: $\zeta<1$ (underdamped), including $\zeta=0$ (undamped)

$$y(t)=1-\frac{1}{\beta}e^{-\zeta\omega_n t}\sin(\beta\omega_n t+\theta),$$

where $\beta=\sqrt{1-\zeta^2}$

$\theta=\tan^{-1}(\beta/\zeta)$

Case 2: $\zeta>1$ (overdamped)

$$y(t)=1+k_1e^{-t/\tau_1}+k_2e^{-t/\tau_2}$$

where $\tau_{1,2}=\dfrac{1}{\zeta\omega_n\pm\omega_n\sqrt{\zeta^2-1}}$

Case 3: $\zeta=1$ (critically damped)

$$y(t)=1+(k_1+k_2t)e^{-t/\tau}$$

where $\tau=1/\omega_n$

Step response of second-order systems

A 2-order system: $G(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

Case 1: $\zeta < 1$ (underdamped),

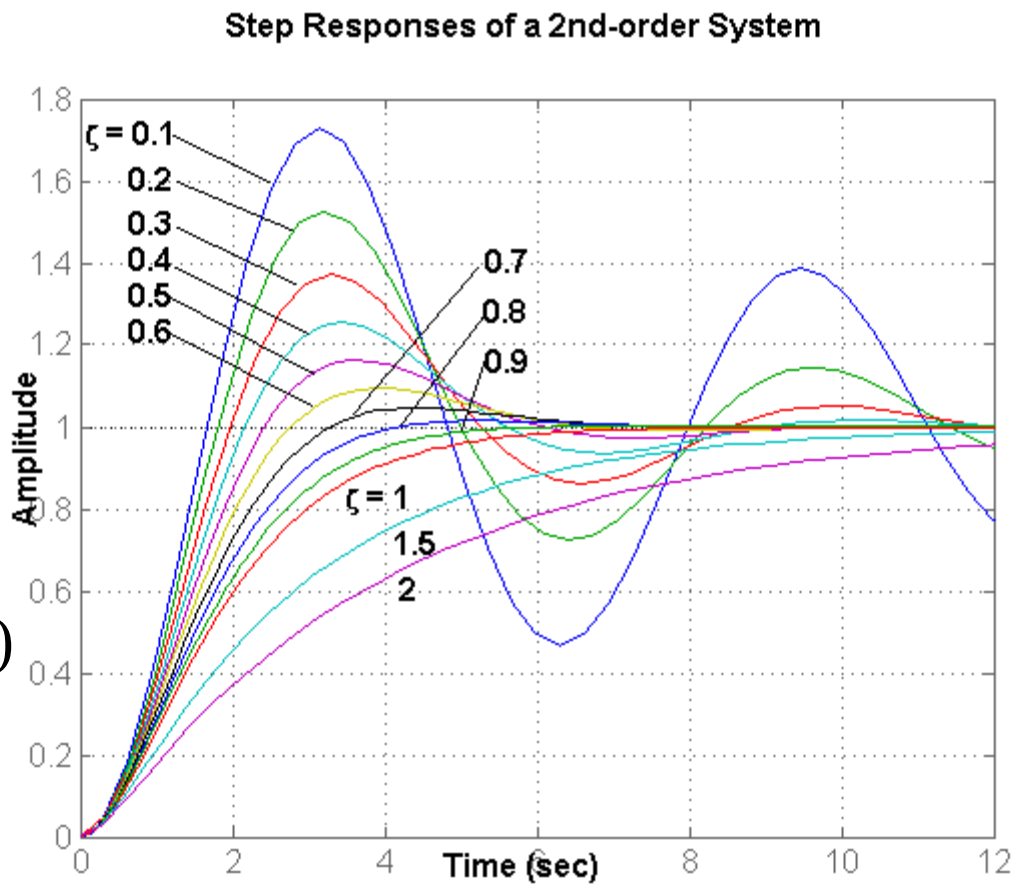
$$y(t) = 1 - \frac{1}{\beta} e^{-\zeta\omega_n t} \sin(\beta\omega_n t + \theta)$$

Case 2: $\zeta > 1$ (overdamped)

$$y(t) = 1 + k_1 e^{-t/\tau_1} + k_2 e^{-t/\tau_2}$$

Case 3: $\zeta = 1$ (critically damped)

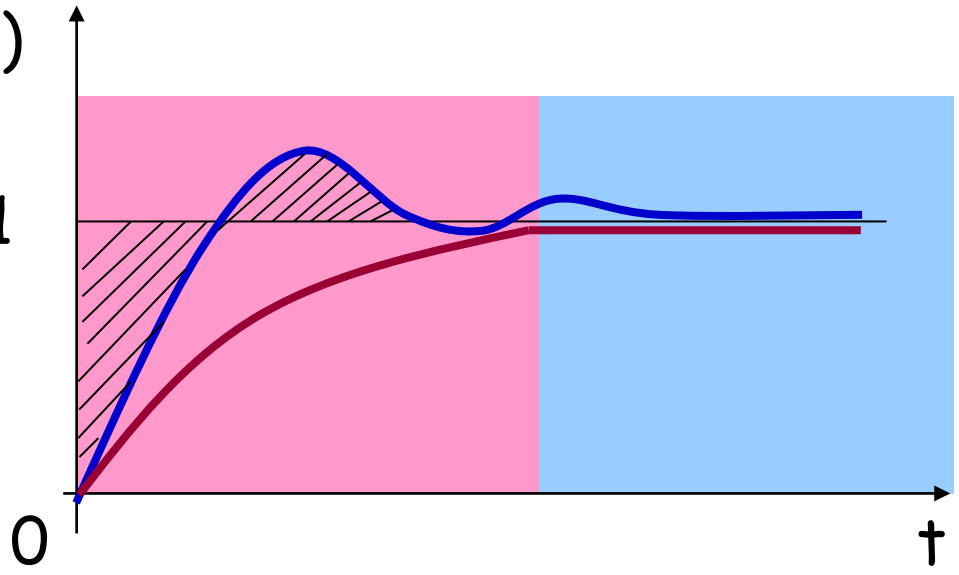
$$y(t) = 1 + (k_1 + k_2 t) e^{-t/\tau}$$



Transient and steady-state response

The time response of a control system is usually divided into two parts:

$$y(t) = y_t(t) + y_{ss}(t)$$



Transient response $y_t(t)$
-- defined as the part of the time response that goes to zero as time becomes very large.

$$\lim_{t\to\infty} y_t(t) = 0$$

Steady-state response $y_{ss}(t)$
-- the part of the time response that remains after the transient has died out.

Time-domain Specifications

1. Steady-state value: y_{ss}
2. Maximum overshoot:

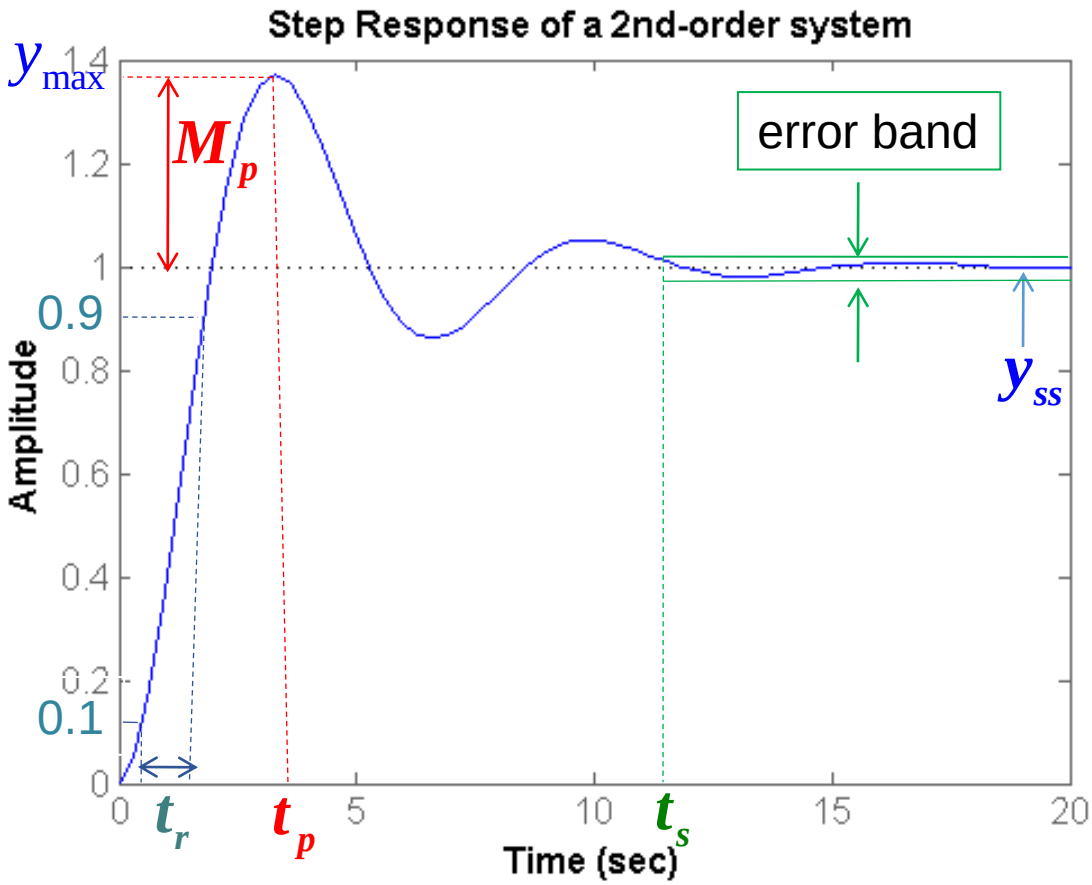
$$M_p = y_{\max} - y_{ss};$$

$$\text{Percent overshoot} = \frac{M_p}{y_{ss}} \times 100\%$$

3. Peak time: t_p

   How to calculate t_p?
4. Rise time: t_r
5. Settling time: t_s

Is there overshoot in the time response of a first-order system?



In practical applications, the following criteria are often used:

Rise time : evaluate the response speed of the system (quickness)

Overshoot: evaluate the damping of the system (smoothness)

Settling time: reflect both response speed and damping

Relationship between the characteristic-equation roots and the step response

A second-order system: $$G(s)=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

Its characteristic equation: $$D(s)=s^2+2\zeta\omega_n s+\omega_n^2=0$$

The value of ζ determines the location of the roots of $D(s)=0$.

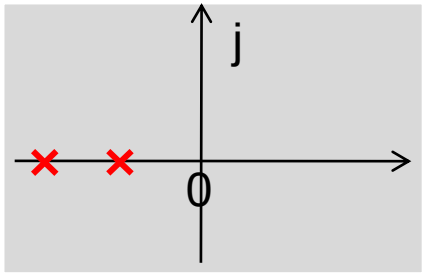
$\zeta>1$:	$s_{1,2}=-\zeta\omega_n\pm\omega_n\sqrt{\zeta^2-1}$	overdamped
$\zeta=1$:	$s_{1,2}=-\omega_n$	critically damped
$0<\zeta<1$:	$s_{1,2}=-\zeta\omega_n\pm j\omega_n\sqrt{1-\zeta^2}$	underdamped
$\zeta=0$:	$s_{1,2}=\pm j\omega_n$	undamped
$\zeta<0$:	$s_{1,2}=-\zeta\omega_n\pm j\omega_n\sqrt{1-\zeta^2}$	negatively damped

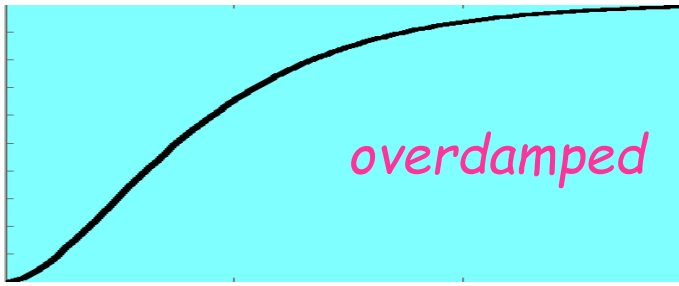
A second-order system:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

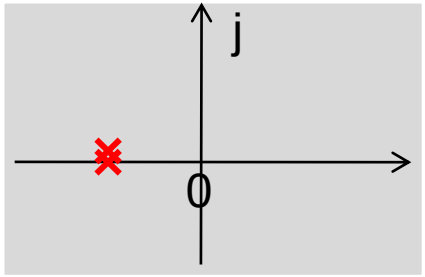
$\zeta > 1$:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$





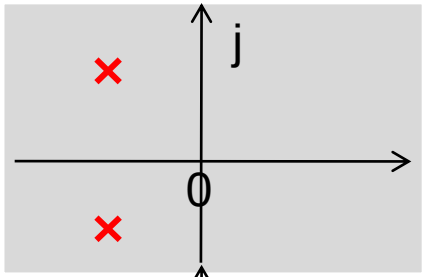
$\zeta = 1$: $s_{1,2} = -\omega_n$

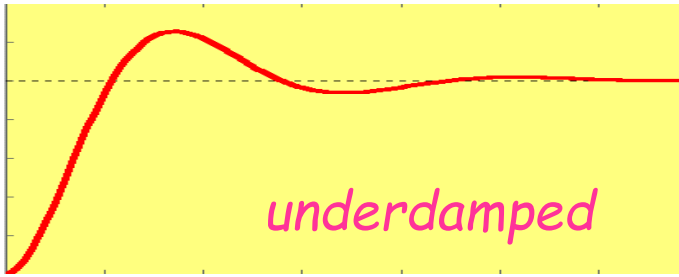




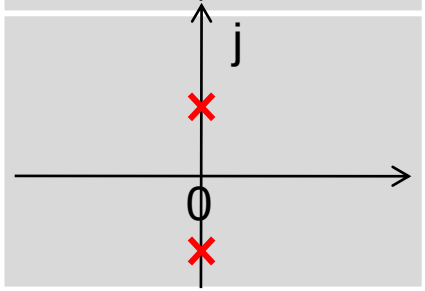
$0 < \zeta < 1$:

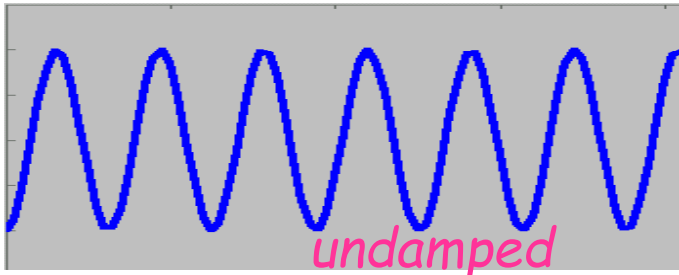
$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$





$\zeta = 0$: $s_{1,2} = \pm j\omega_n$





A second-order system:

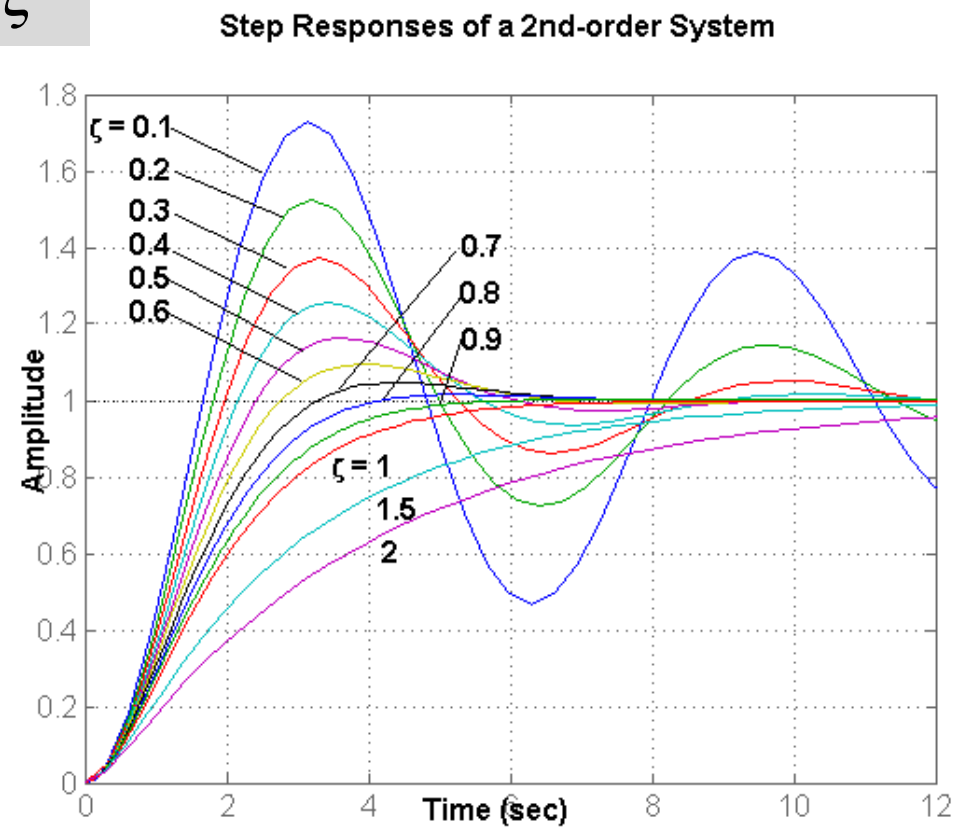
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Effects of damping ratio ζ

(for a given ω_n)

ζ ↑ ⇨

- **Overshoot M_p** ↓
  The oscillation is smaller
- **rise time t_r** ↑
  The speed of the response is slower



We are confronted with a necessary compromise between the speed of response and the allowable overshoot.

Note: In control engineering, except those systems that do not allow any oscillation, usually a control system is designed with

- moderate damping (allowing some overshoot)
- quick response speed
- short settling time

Therefore, a second-order control system is usually designed as an underdamped system.

Underdamped second-order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ζ, damping ratio

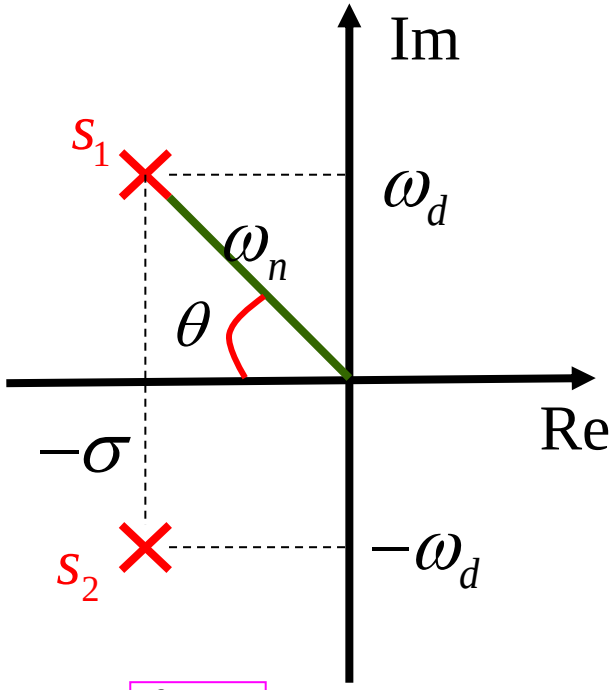
ω_n, natural undamped frequency

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$= -\sigma \pm j\omega_d$$

$\sigma = \zeta\omega_n$, dampting factor

$\omega_d = \omega_n\sqrt{1-\zeta^2}$, damped frequency



$\theta = ?$

$$\theta = \arccos\zeta$$

$$\theta = \arctan\frac{\sqrt{1-\zeta^2}}{\zeta}$$

Performance analysis

Unit-step response:

$$y(t)=1-\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t+\theta),\ 0<\zeta<1$$

1. Rise Time

$$y(t_r)=1,\quad \text{that is,}\quad \frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}}\cdot\sin(\omega_d t_r+\theta)=0$$

$$\frac{e^{-\zeta\cdot\omega_n\cdot t_r}}{\sqrt{1-\zeta^2}}\neq 0,\quad \Rightarrow \omega_d t_r+\theta=n\pi(n=0,\pm1,\pm2,\cdots)$$

$$t_r=\frac{\pi-\theta}{\omega_d}=\frac{\pi-\theta}{\omega_n\sqrt{1-\zeta^2}}$$

t_r is the time needed for the response to reach the steady-state value for the first time, so n=1.

For a given ω_n $\zeta\downarrow$, $t_r\downarrow$;
For a given ζ, $\omega_n\uparrow$, $t_r\downarrow$.

$$y(t)=1-\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t+\theta),\ 0<\zeta<1$$

$$\frac{dy(t)}{dt}=0$$

2 . Peak time

$$\frac{dy(t)}{dt}=\left(\frac{\zeta^2\omega_n}{\sqrt{1-\zeta^2}}+\omega_n\sqrt{1-\zeta^2}\right)e^{-\zeta\cdot\omega_n\cdot t}\sin\omega_d t$$

$$=\frac{\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\cdot\omega_n\cdot t}\sin\omega_d t\ =0$$

$$\therefore \sin\omega_d t_p=0\Rightarrow\omega_d t_p=n\pi(n=0,\pm1,\pm2,\cdots)$$

t_p is the time needed for the response to reach the maximum value for the first time, so n=1.

$$\omega_d t_p=\pi\quad\Rightarrow\quad t_p=\frac{\pi}{\omega_d}=\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

For a given ω_n, $\zeta\downarrow$, $t_p\downarrow$;
For a given ζ, $\omega_n\uparrow$, $t_p\downarrow$.

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta), 0 < \zeta < 1$$

$$t_p = \frac{\pi}{w_d} = \frac{\pi}{w_n\sqrt{1-\zeta^2}}$$

3. Overshoot

$$M_p\% = \frac{y(t_p) - y(\infty)}{y(\infty)} \times 100\%$$

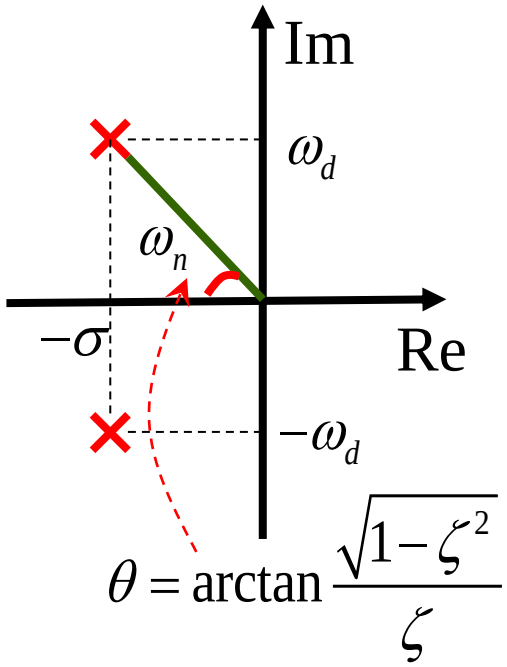
$$y(t_p) = 1 - \frac{e^{-\zeta \cdot w_n \cdot t_p}}{\sqrt{1-\zeta^2}} \sin(\pi + \theta)$$

$$\sin(\pi + \theta) = -\sin\theta = -\sqrt{1-\zeta^2}$$

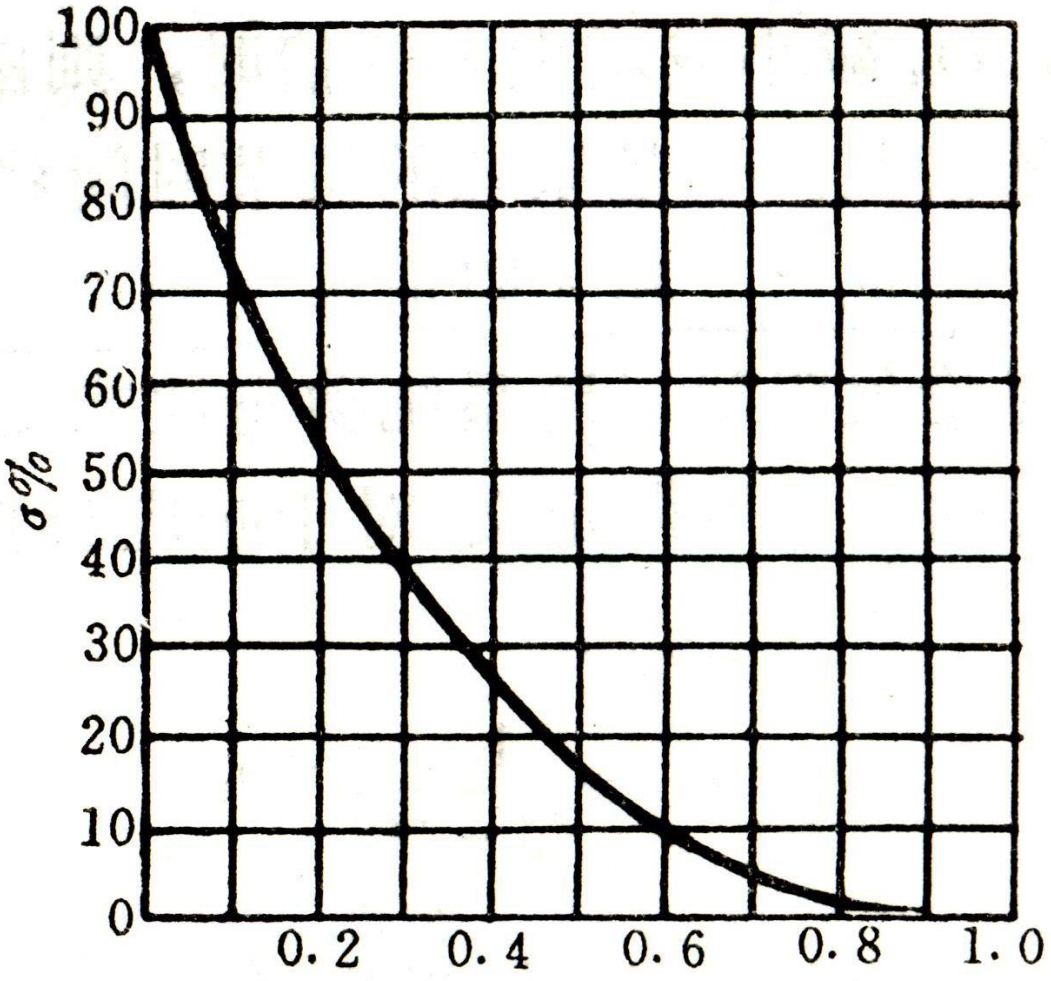
$$\therefore y(t_p) = 1 + e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

Note that $y(\infty) = 1$

Thus $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$



Overshoot is a function of damping ratio ζ, independent of ω_n.



$\zeta \uparrow$, $M_p\% \downarrow$

Usually, ζ is set between 0.4 and 0.8 to get better smoothness and quickness, the corresponding overshoot is between 25% and 2.5%.

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta), 0 < \zeta < 1$$

4 Settling Time

With the definition of error band

$$\left| \frac{e^{-\zeta \cdot \omega_n \cdot t}}{\sqrt{1-\zeta^2}} \cdot \sin(\sqrt{1-\zeta^2} \cdot \omega_n t_s + \theta) \right| \leq 0.05 \text{ or } 0.02$$

t_s cannot be obtained directly, but we can get the relationship between $\zeta\omega_n$ and t_s.

Relationship between (ζ, ω_n) and (t_r, t_p, M_p, t_s)

$$t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - \theta}{\omega_n \sqrt{1-\zeta^2}}$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

For a given ω_n

$\zeta \downarrow \Rightarrow t_r \downarrow t_p \downarrow$

$\zeta \downarrow \Rightarrow M_p \uparrow$

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

ζ can be calculated by the requirement on the overshoot M_p. $\zeta \in [0.4, 0.8]$.

$$t_s \approx \frac{3}{\zeta\omega_n} = \frac{3}{\sigma} = 3\tau(\Delta = 5\%)$$

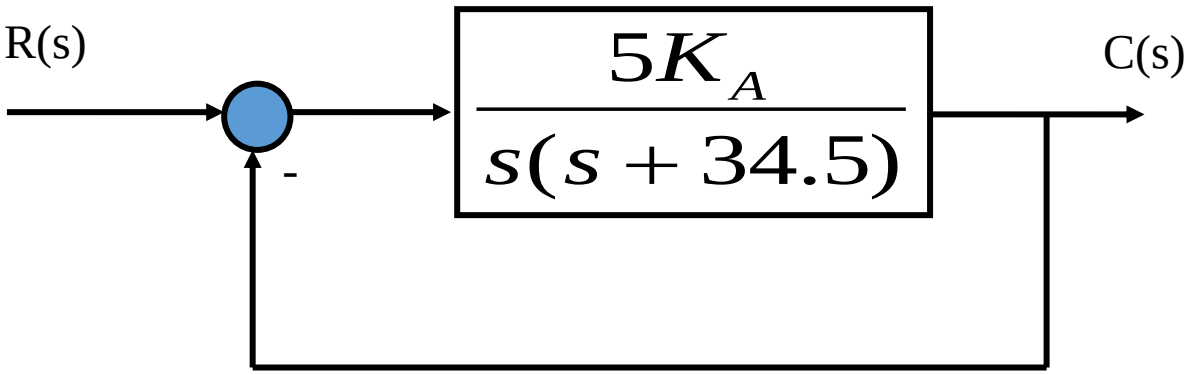
$$t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{\sigma} = 4\tau(\Delta = 2\%)$$

$$t_s \approx \frac{5}{\zeta\omega_n} = \frac{5}{\sigma} = 5\tau(\Delta = 1\%)$$

$\zeta\omega_n \uparrow \Rightarrow t_s \downarrow$

Once ζ is determined, ω_n can be determined based on the requirement on error band $\Delta\%$.

Example 1: Consider the following unit-feedback system



System input is the unit-step function. When the amplifier gains are K_A=200, K_A=1500, K_A=13.5 respectively, calculate the time-domain specifications of the unit-step response.

Investigate the effect of the amplifier gain K_A on the system response

Solution: The closed-loop transfer function is

$$T(s) = \frac{G(s)}{1+G(s)} = \frac{5K_A}{s^2 + 34.5s + 5K_A}$$

$$K_A = 200, \therefore T(s) = \frac{1000}{s^2 + 34.5s + 1000}$$

$$\therefore \omega_n^2 = 1000, \quad 2\zeta\omega_n = 34.5$$

$$\therefore \omega_n = 31.6(rad/s), \zeta = \frac{34.5}{2\omega_n} = 0.545$$

According to the formula to calculate the performance indices, it follows that

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = 0.12(\text{sec})$$

$$t_s \approx \frac{3}{\zeta \omega_n} = 0.174(\text{sec})$$

$$M_p\% = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = 13\%$$

$K_A = 1500$

If $K_A = 200$, then $\omega_n = 34.5(rad/s)$; $\zeta = 0.545$

$\therefore t_p = 0.12(s)$, $t_s = 0.174(s)$, $M_p\% = 13\%$

If $K_A = 1500$, then $\omega_n = 86.2(rad/s)$; $\zeta = 0.2$

$\therefore t_p = 0.037(s)$, $t_s = 0.174(s)$, $M_p\% = 52.7\%$

Thus, the greater the K_A, the less the ζ, the greater the w_n, the less the t_p, the greater the $M_p\%$, while the settling time t_s has no change.

$K_A = 13.5$

$\zeta > 1$ Overdamped

When $K_A = 13.5, \omega_n = 8.22(rad/s), \zeta = 2.1$

When $K_A = 13.5, \omega_n = 8.22(rad/s), \zeta = 2.1$

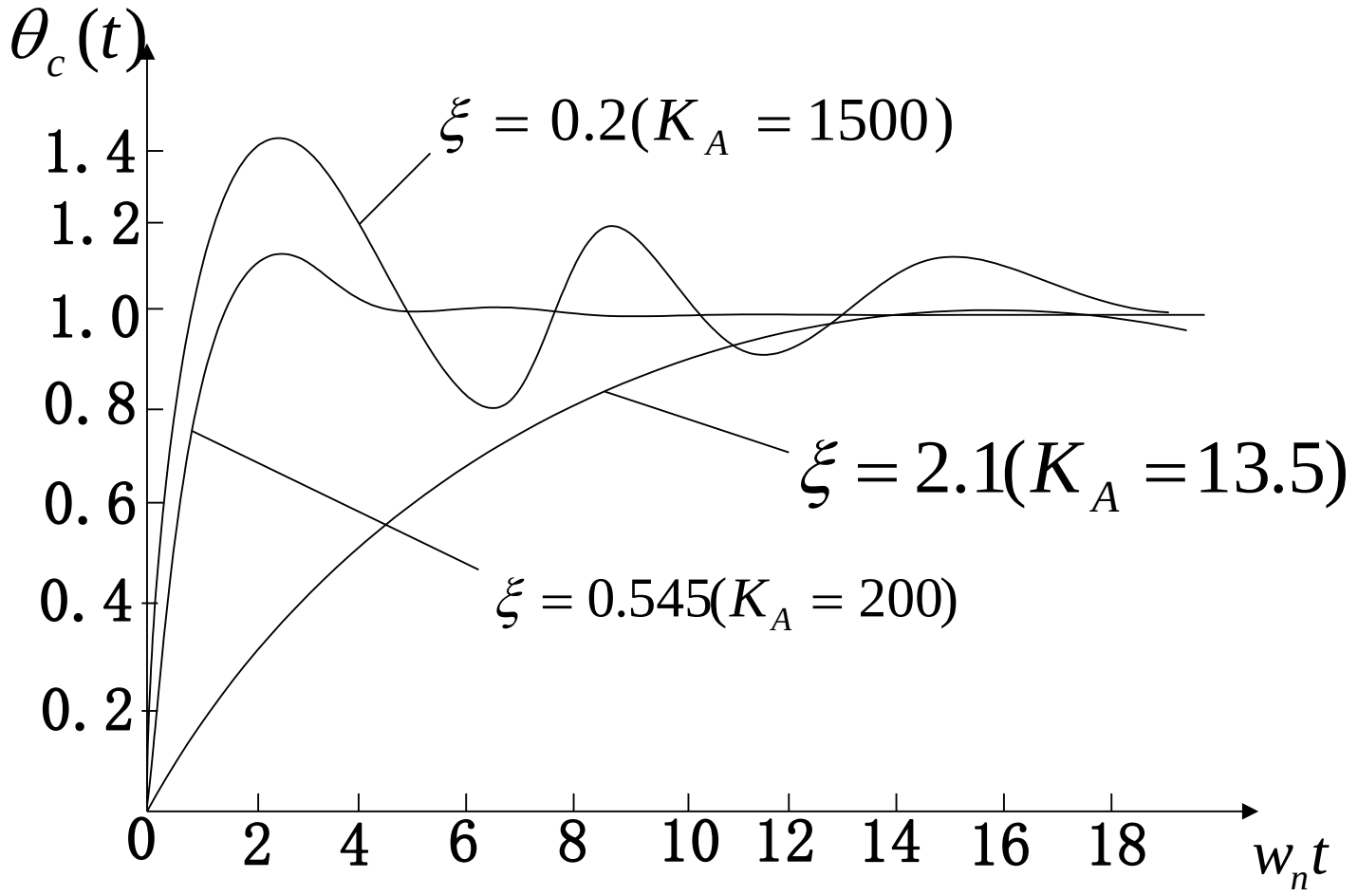
When system is over-damped, there is no peak time, overshoot and oscillation.

The settling time can be calculated approximately:

$$t_s \approx 3\tau_1 = 1.46(\text{sec})$$

$$\frac{1}{\tau_1} = \omega_n(\zeta - \sqrt{\zeta^2 - 1})$$

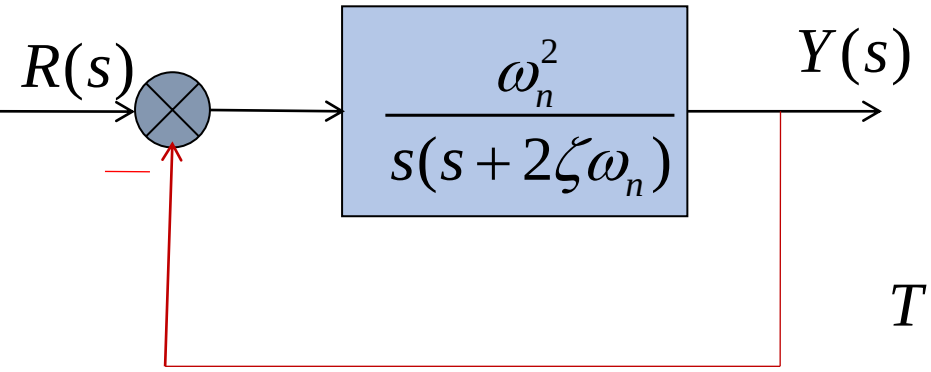
The settling time is greater than previous cases, although the response has no overshoot, the transition process is very slow, the curves are as follows:



Note: When K_A increases, t_p decreases, t_r decreases, the speed of response increases, meanwhile, the overshoot increases. Therefore, to improve the dynamic performance indexes of system, we adopt PD-control or velocity feedback control, namely, PD compensation.

Effects of Adding Poles and Zeros to Transfer Functions

Effects of Adding Poles

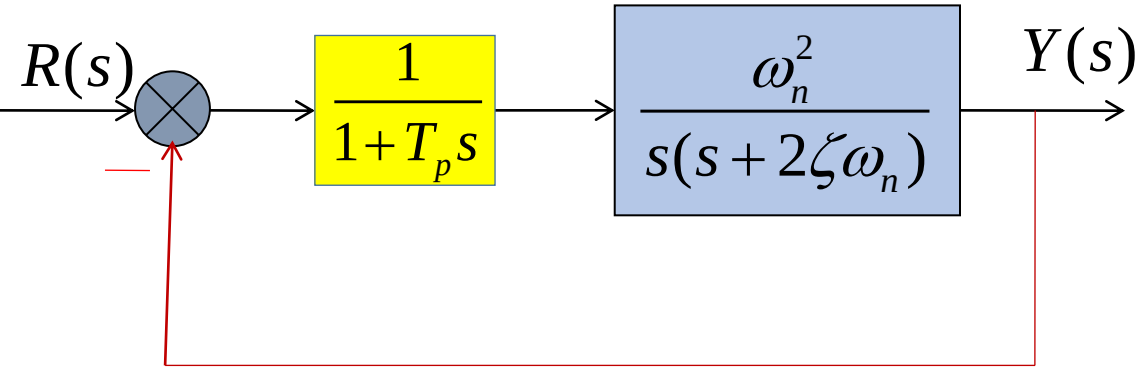


Open-loop TF: $G(s) = \dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$

Closed-loop TF:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

1. Adding a pole at $s = -1/T_p$ to the open-loop TF



Open-loop TF:

$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)(1+T_p s)}$$

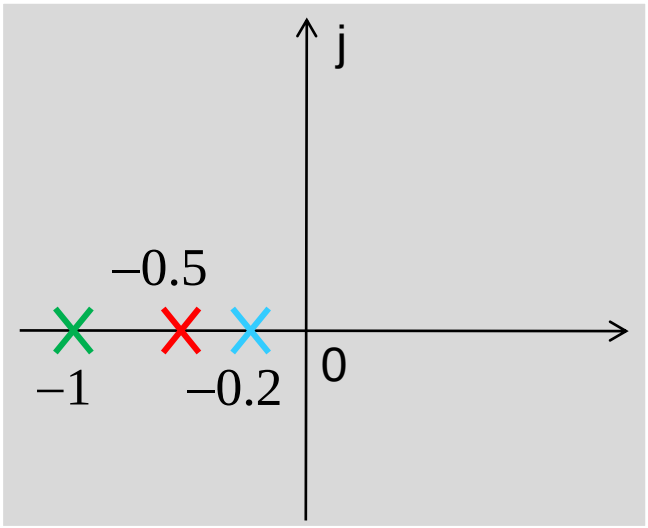
Closed-loop TF: $T(s) = \dfrac{\omega_n^2}{T_p s^3 + (1+2\zeta\omega_n T_p)s^2 + 2\zeta\omega_n s + \omega_n^2}$

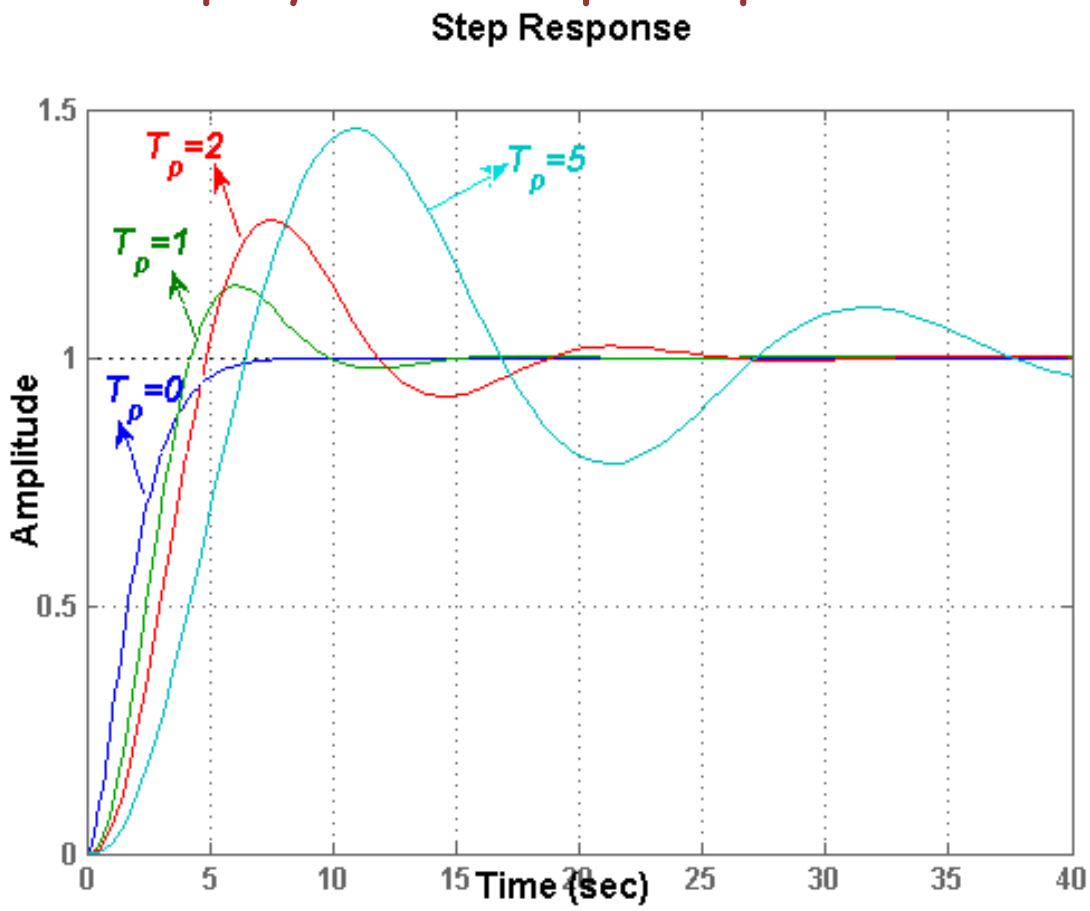
1. Adding a pole at $s = -1/T_p$ to the open-loop TF

How does it affect closed-loop system step-response performance?

$T_p = 0,\ 1,\ 2,\ 5$

$poles:\ s = \infty,\ -1,\ -0.5,\ -0.2$

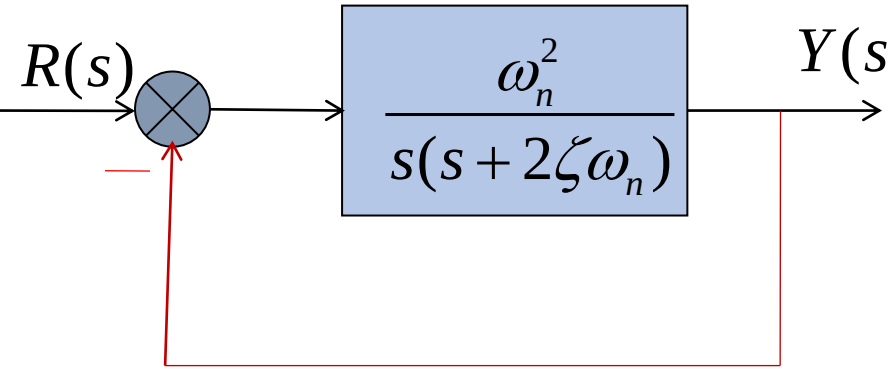




-- Increasing the maximum overshoot of the closed-loop system;

-- Increasing the rise time of the closed-loop system.

Effects of Adding Poles



Open-loop TF: $G(s) = \dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$

Closed-loop TF:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

2. Adding a pole at $s = -1/T_p$ to the closed-loop TF

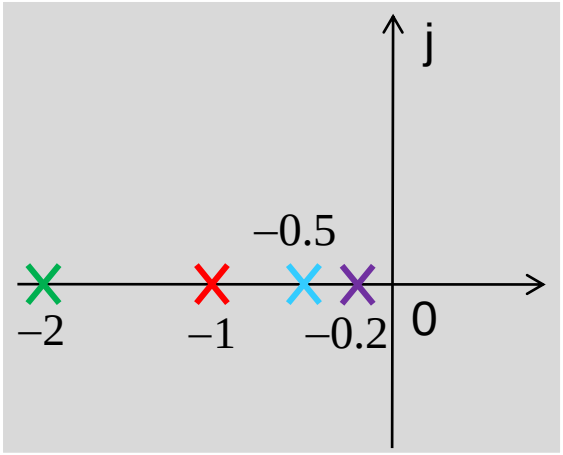
Closed-loop TF: $$T(s) = \frac{\omega_n^2}{(s^2+2\zeta\omega_n s+\omega_n^2)(1+T_p s)}$$

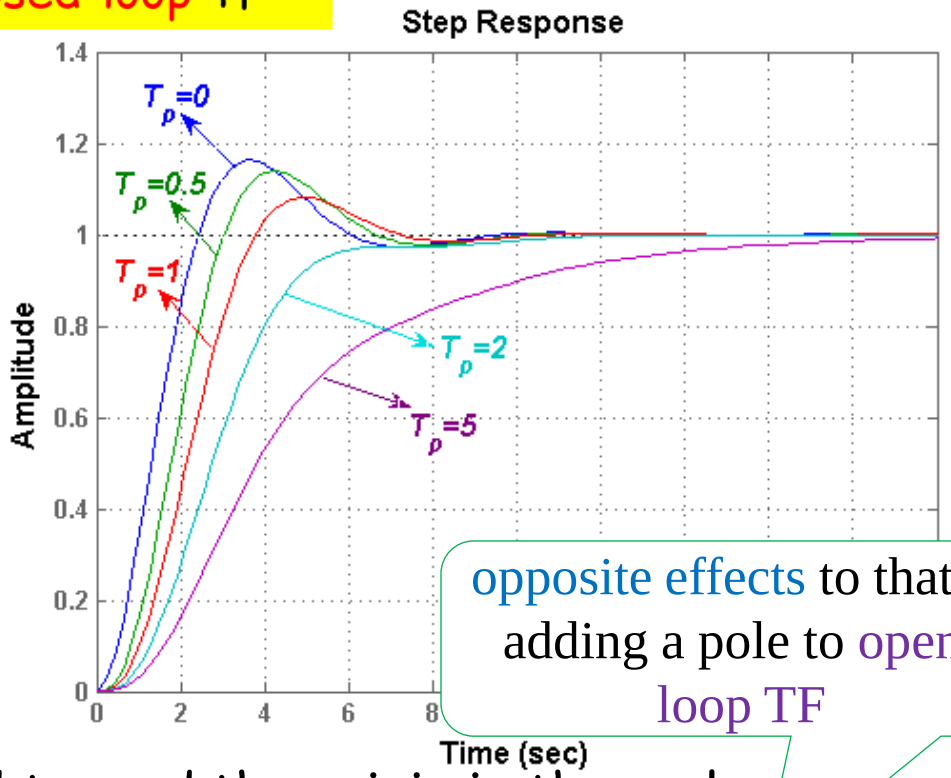
$$= \frac{\omega_n^2}{T_p s^3+(1+2\zeta\omega_n T_p)s^2+(2\zeta\omega_n+\omega_n^2 T_p)s+\omega_n^2}$$

2. Adding a pole at $s = -1/T_p$ to the closed-loop TF

$T_p = 0,\ 0.5,\ 1,\ 2,\ 5$

$s = \infty,\ -2,\ -1,\ -0.5,\ -0.2$





opposite effects to that of adding a pole to open-loop TF

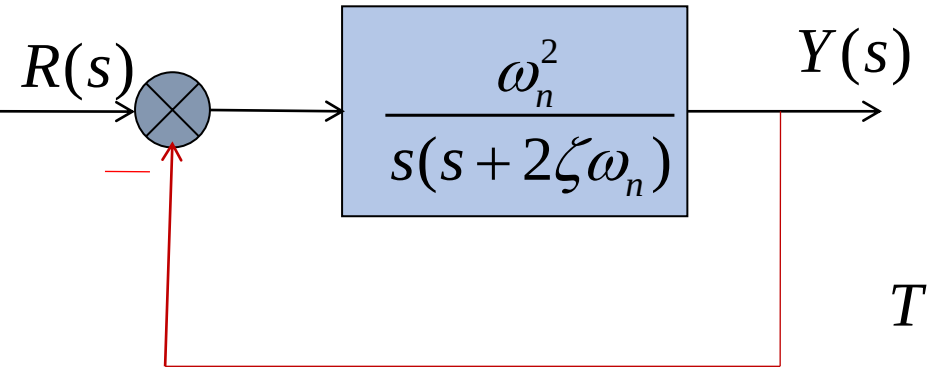
As the pole at $s = -1/T_p$ is moved toward the origin in the s-plan

-- the maximum overshoot of the closed-loop system decreases;

-- the rise time of the closed-loop system increases.

1. Adding a pole to the closed-loop system has the effect similar to increasing the damping ratio;
2. An originally underdamped system can be made overdamped by adding a closed-loop pole.

Effects of Adding Zeros



Open-loop TF: $G(s)=\dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$

Closed-loop TF:

$$T(s)=\frac{Y(s)}{R(s)}=\frac{G(s)}{1+G(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

1. Adding a zero at $s=-1/T_z$ to the closed-loop TF

Closed-loop TF:

$$T(s)=\frac{Y(s)}{R(s)}=\frac{\omega_n^2(1+T_z s)}{s^2+2\zeta\omega_n s+\omega_n^2}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}+\frac{\omega_n^2 T_z s}{s^2+2\zeta\omega_n s+\omega_n^2}$$

1. Adding a zero at $s=-1/T_z$ to the closed-loop TF

Closed-loop TF:

$$T(s)=\frac{Y(s)}{R(s)}=\frac{\omega_n^2(1+T_z s)}{s^2+2\zeta\omega_n s+\omega_n^2}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}+\frac{\omega_n^2 T_z s}{s^2+2\zeta\omega_n s+\omega_n^2}$$

For a unit-step input $1(t)$, $R(s)=\frac{1}{s}$

The step response of the closed-loop system

$$Y(s)=T(s)R(s)=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}\cdot\frac{1}{s}+T_z s\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}\cdot\frac{1}{s}$$

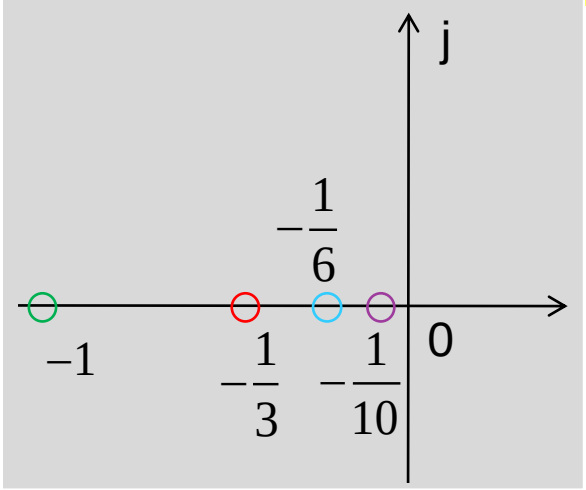
$$y(t)=L^{-1}[Y(s)]=y_1(t)+T_z\frac{dy_1(t)}{dt}$$

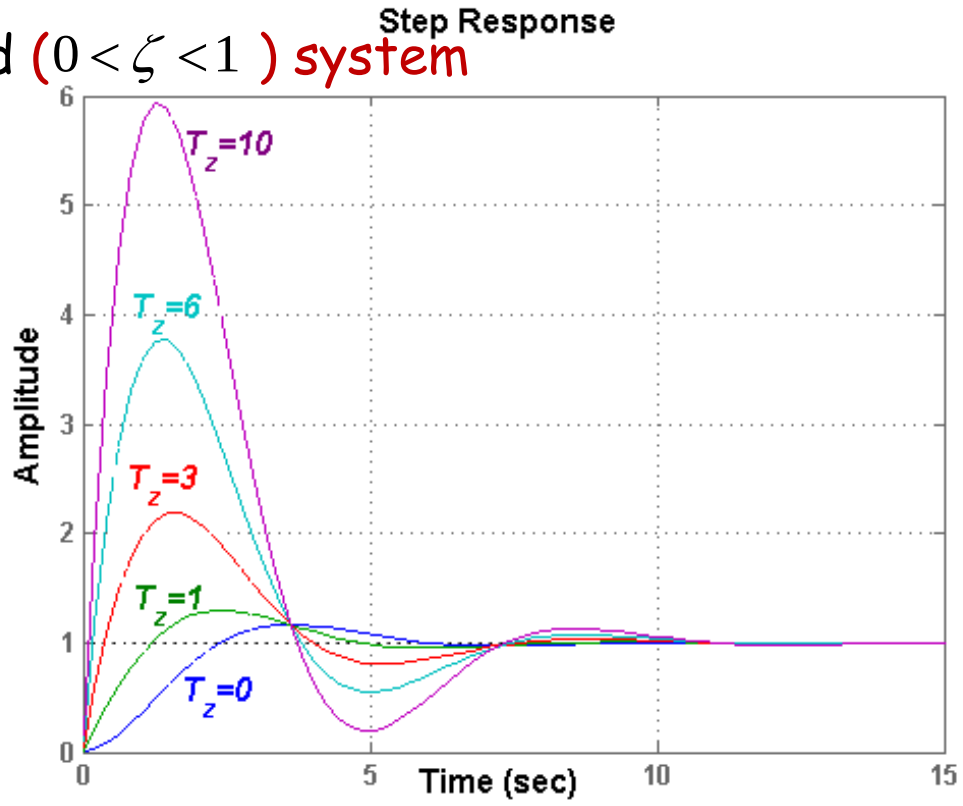
1. Adding a zero at $s = -1/T_z$ to the closed-loop TF

a) Its effects on an underdamped ($0 < \zeta < 1$) system

$T_Z = 0,\ 1,\ 3,\ 6,\ 10$

$s = \infty,\ -1,\ -\frac{1}{3},\ -\frac{1}{6},\ -\frac{1}{10}$





As the zero at $s = -1/T_z$ is moved toward the origin in the s-plane

-- the maximum overshoot of the closed-loop system increases;

-- the rise time of the closed-loop system decreases.

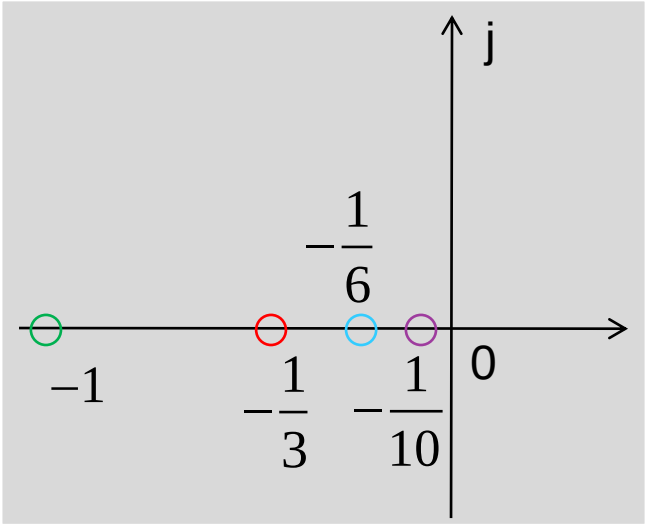
The effect is similar to the effect of reducing the damping ratio

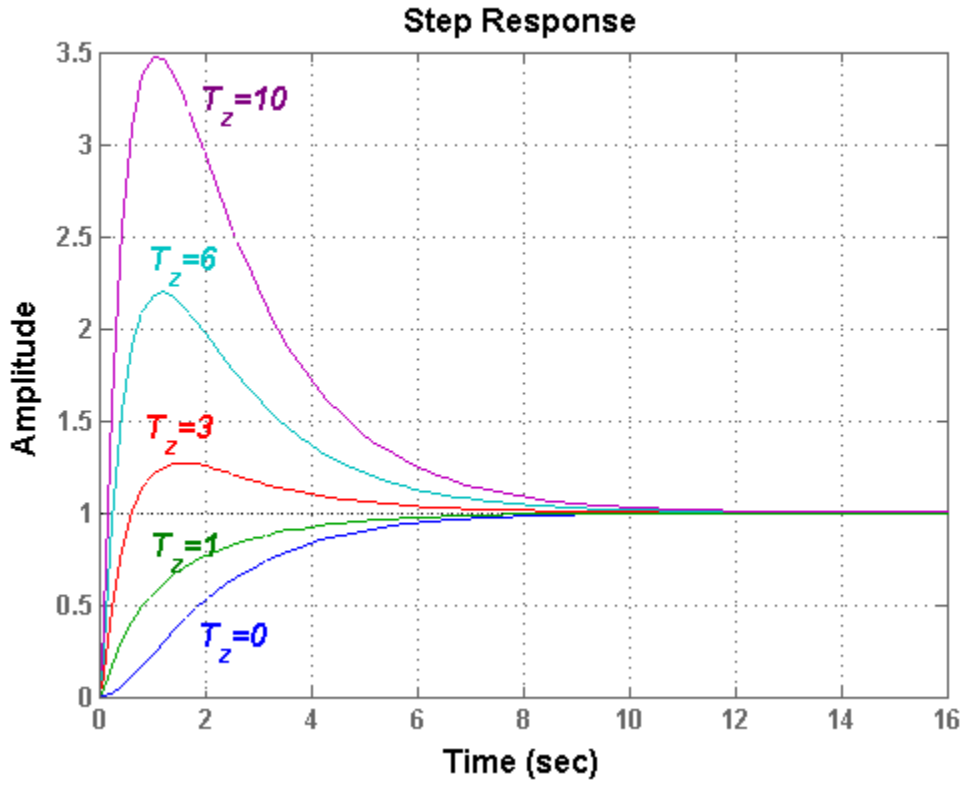
1. Adding a zero at $s = -1/T_z$ to the closed-loop TF

b) Its effects on an overdamped ($\zeta > 1$) system

$T_Z = 0,\ 1,\ 3,\ 6,\ 10$

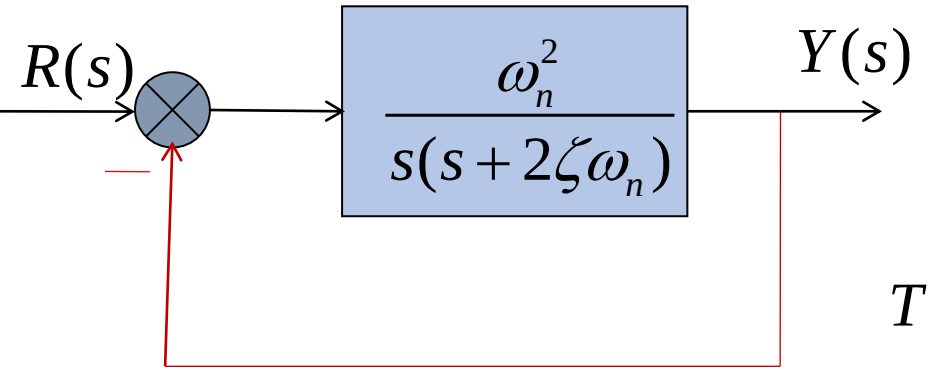
$s = \infty,\ -1,\ -\frac{1}{3},\ -\frac{1}{6},\ -\frac{1}{10}$





Adding a zero to an overdamped system can result in overshoot (similar to an underdamped system). This is only one time overshoot.

Effects of Adding Zeros

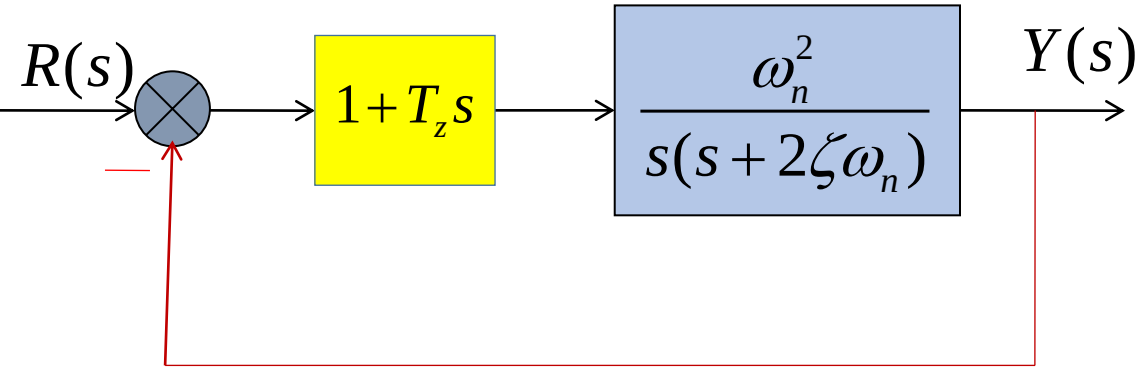


Open-loop TF: $G(s) = \dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$

Closed-loop TF:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

1. Adding a zero at $s = -1/T_z$ to the open-loop TF



Open-loop TF:

$$G(s) = \frac{(1+T_z s)\omega_n^2}{s(s+2\zeta\omega_n)}$$

Closed-loop TF: $T(s) = \dfrac{\omega_n^2(1+T_z s)}{s^2+(2\zeta\omega_n+T_z\omega_n^2)s+\omega_n^2}$

The additional zero changes both numerator and denominator.

1. Adding a zero at $s=-1/T_z$ to the open-loop TF

The additional zero changes both numerator and denominator.

Closed-loop TF:

$$T(s)=\frac{\omega_n^2(1+T_z s)}{s^2+(2\zeta\omega_n+T_z\omega_n^2)s+\omega_n^2}=\frac{\omega_n^2(1+T_z s)}{s^2+2\zeta_z\omega_n s+\omega_n^2}$$

A closed loop additional zero will increase overshoot

$s=-1/T_z$

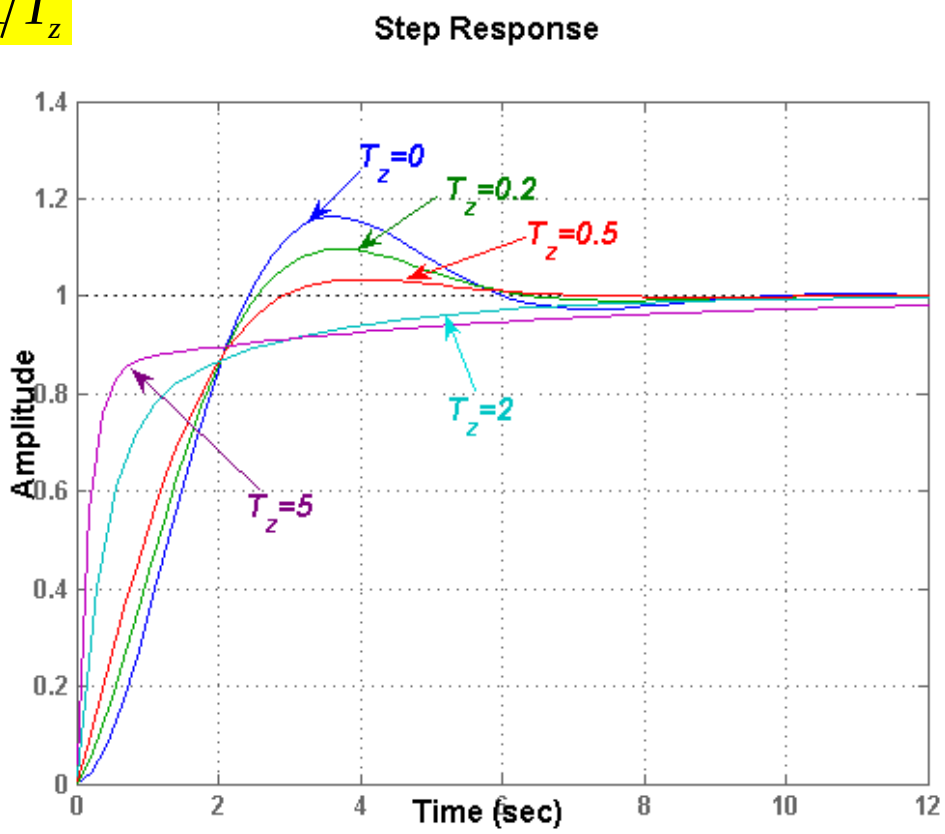
$T_z\uparrow$ overshoot $M_p\%\uparrow$

The equivalent damping ratio will decrease the overshoot:

$$\zeta_z=\zeta+\frac{T_z\omega_n}{2}>\zeta$$

$T_z\uparrow$ $\zeta_z\uparrow$ overshoot $M_p\%\downarrow$

when $\zeta_z>1$, the closed-loop system becomes overdamped (no overshoot no matter how large T_z is)



Dominant Poles of Transfer Function

Dominant poles: those poses that have a dominant effect on the transient response.

By identifying dominant poles, high-order systems can be approximated by lower ones as the transient response is concerned.

e.g.

$$Y(s) = \frac{1}{(s+p_1)(s+p_2)(s+p_3)} = \frac{c_1}{s+p_1} + \frac{c_2}{s+p_3} + \frac{c_3}{s+p_3}$$

$$y(t) = c_1 e^{-p_1 t} + c_2 e^{-p_2 t} + c_3 e^{-p_3 t}$$

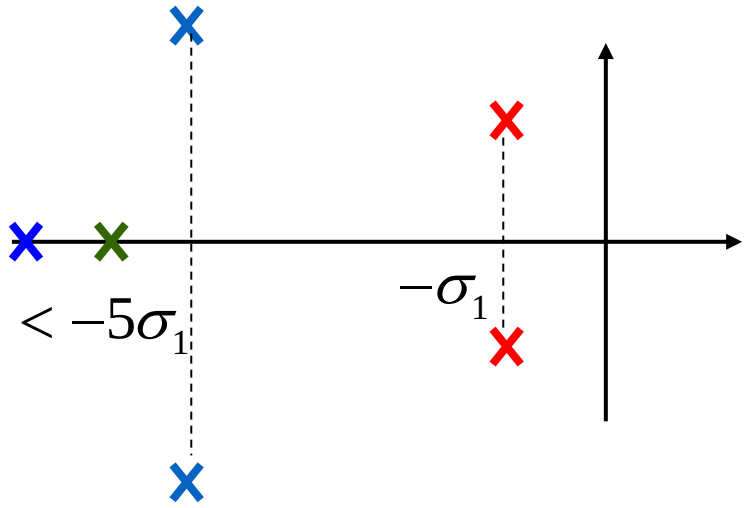
poles: $-p_1, -p_2, -p_3$

If $p_1 > p_2 > p_3$, $c_1 e^{-p_1 t}$ decays fastest, $c_3 e^{-p_3 t}$ decays slowest.

Position of Poles in the left-half s-plane	Their effects on transient response
close to the imaginary axis	decaying relatively slowly
far away from the imaginary axis	decaying fast

Dominant Poles of Transfer Function

If the ratio of real parts exceed 5 and no zeros nearby, the closed-loop poles nearest to the imaginary-axis will dominate in the transient response behavior.



A real pole can be a dominant pole, but a pair of complex conjugate poles are usually more preferable in control engineering (why?) .

In order to apply second-order system in approximating the dynamic performance of higher-order system

Stability of linear systems (in complex plane)

The concept of stability

Two types of response for LTI systems:

--Zero-state response: the response is due to input only; all the initial conditions are zero;

Bounded-input-bounded-output (BIBO) stability: with zero initial conditions, for every bounded input u(t), the output y(t) of the system is bounded.

--Zero-input response: the response is due to the initial conditions only; all the inputs are zero;

Asymptotic stability: with zero input, for every finite initial conditions $y(t_0), \dot{y}(t_0), \cdots, y^{(n)}(t_0)$, if there exist a positive number M which depends on the initial conditions, such that

$$(1)\quad |y(t)| \le M < \infty \quad \textit{for all } t > t_0; \quad \text{and} \quad (2)\quad \lim_{t\to\infty} |y(t)| = 0.$$

then the LTI system is asymptotically stable.

Time-domain condition for stability:

Assume that the initial condition of the system is zero. Let $g(t)$ be the *impulse response* of the system. In other words, when system input is unit impulse function $\delta(t)$, the system output is $g(t)$.

If $\int_0^\infty |g(t)|dt \le M < \infty$, then the system is BIBO stable.

This usually holds when $\lim_{t\to\infty} g(t) = 0$

$$G(s) = \frac{Y(s)}{U(s)} = \sum_{i=1}^{n} \frac{C_i}{s + p_i} \qquad g(t) = L^{-1}[G(s)] = \sum_{i=1}^{n} C_i e^{-p_i t}$$

$$\lim_{t\to\infty} g(t) = 0 \quad \Rightarrow e^{-p_i t} \text{ decay with time}$$

All poles should be located in the left side of s-plane

Stability condition in complex plane

A system is stable if and only if

all roots of the closed-loop system characteristic equation have negative real parts

or equivalently

all poles of closed-loop transfer functions are located in the open left half of the s-plane.

For LTI systems, both BIBO stability and asymptotic stability have the same requirement on pole location. Thus, if a system is BIBO stable, it is also asymptotically stable.

So, we simply refer to the stability condition of a LTI system as stable or unstable.

LTI systems

Stability Conditions	Location of poles
stable	all poles in LHP
marginally stable	simple poles on the jw-axis and no poles in RHF
unstable	at least one pole in RHF or at least one multi-order (i.e., repeated) pole on the jw-axis

Routh-Hurwitz stability criterion

All poles in left s-plane => No poles in right s-plane

- The criterion tests whether any of the roots of the characteristic equation lie in the right half of the s-plane, without actually calculating the roots.
- Information about stability can be obtained directly from the coefficients of the characteristic equation

Routh-Hurwitz's stability criterion

A necessary (but not sufficient) condition for stability:

(1) All the coefficients of the characteristic equation have the same sign.
(2) None of the coefficients vanishes.

Consider the characteristic equation of a LTI system

$$D(s) = a_0 s^n + a_1 s^{n-1} + \ldots + a_{n-1} s + a_n = 0, \quad a_0 > 0$$

Note 1: The above conditions are based on the laws of algebra .

$$(s-s_1)(s-s_2)(s-s_3)=0 \rightarrow a_0 s^3 + a_1 s^2 + a_2 s + a_3 = 0$$

$$s(s-s_2)(s-s_3) - s_1(s-s_2)(s-s_3) = 0$$

$$s(s^2 - (s_2+s_3)s + s_2 s_3) - s_1(s^2 - (s_2+s_3)s + s_2 s_3) = 0$$

$$s^3 - (s_1+s_2+s_3)s^2 + (s_1 s_2 + s_1 s_3 + s_2 s_3)s - s_1 s_2 s_3 = 0$$

$$\frac{a_1}{a_0} = -\sum_{i=1}^{3} s_i > 0 \qquad \frac{a_2}{a_0} = \sum_{\substack{i,j=1 \\ i \neq j}}^{3} s_i s_j > 0 \qquad \frac{a_3}{a_0} = -\sum_{\substack{i,j,k=1 \\ i \neq j \neq k}}^{3} s_i s_j s_k > 0$$

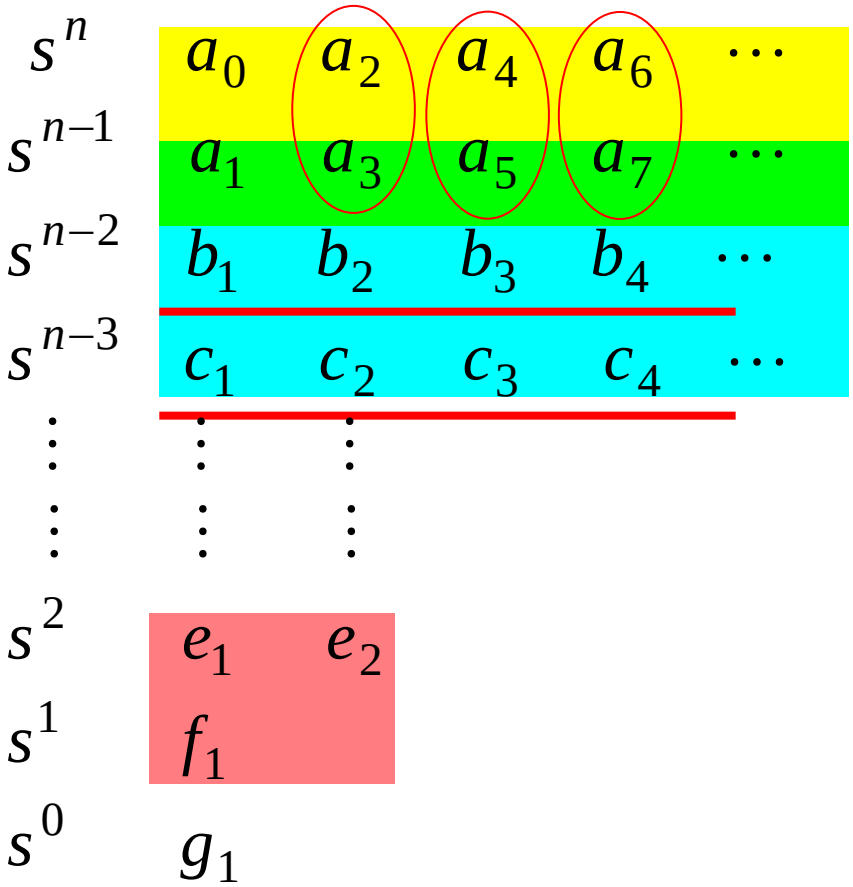
If all roots of the system characteristic equation have negative real parts, all the coefficients have the same sign

Note 2: These conditions are not sufficient.

Routh's Tabulation

Consider the characteristic equation of a LTI system

$$D(s) = a_0 s^n + a_1 s^{n-1} + \ldots + a_{n-1} s + a_n = 0, \quad a_0 > 0$$



$$b_1 = \frac{-1}{a_1}\begin{vmatrix} a_0 & a_2 \\ a_1 & a_3 \end{vmatrix} \qquad c_1 = \frac{-1}{b_1}\begin{vmatrix} a_1 & a_3 \\ b_1 & b_2 \end{vmatrix}$$

$$b_2 = \frac{-1}{a_1}\begin{vmatrix} a_0 & a_4 \\ a_1 & a_5 \end{vmatrix} \qquad c_2 = \frac{-1}{b_1}\begin{vmatrix} a_1 & a_5 \\ b_1 & b_3 \end{vmatrix}$$

$$b_3 = \frac{-1}{a_1}\begin{vmatrix} a_0 & a_6 \\ a_1 & a_7 \end{vmatrix} \qquad c_3 = \frac{-1}{b_1}\begin{vmatrix} a_1 & a_7 \\ b_1 & b_4 \end{vmatrix}$$

Routh's Criterion

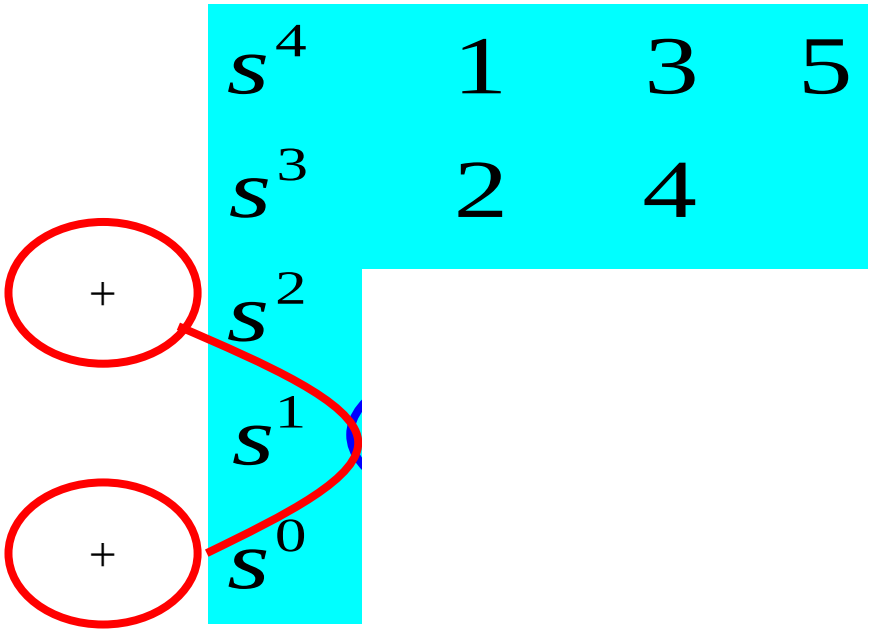
A necessary and sufficient condition for stability:

all the elements of the first column of the Routh' Tabulation are of the same sign.

The number of changes of signs in the elements of the first column equals the number of roots in the right-half s-plane.

Example

$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$



Therefore, the system is unstable and has two roots in the right-half s-plane.

First-order: $a_0 s + a_1 = 0$

If a_0 and a_1 have the same sign, the system is stable.

Second-order: $a_0 s^2 + a_1 s + a_2 = 0$

If a_0 , a_1 and a_2 have the same sign, the system is stable.

Third-order:

$$a_0 s^3 + a_1 s^2 + a_2 s + a_3 = 0$$

s^3	a_0	a_2
s^2	a_1	a_3
s^1	$-\dfrac{a_0 a_3 - a_1 a_2}{a_1}$	
s^0	a_3	

If a_0, a_1, a_2, a_3 are all positive and
$a_1 a_2 > a_3 a_0$
the system is stable.

Special cases when applying Routh's Tabulation

Case 1: only the first element in one of the rows of Routh's tabulation is zero

Solution: replace the zero with a small positive constant ε and proceed as before by taking the limit as $\varepsilon \to 0$

$$s^4 + 3s^3 + 4s^2 + 12s + 16 = 0$$

s^4	1	4	16
s^3	3	12	
s^2	$0(\varepsilon)$	16	
s^1	$\frac{12\varepsilon - 48}{\varepsilon}$	0	
s^0	16		

when $\varepsilon \to 0$

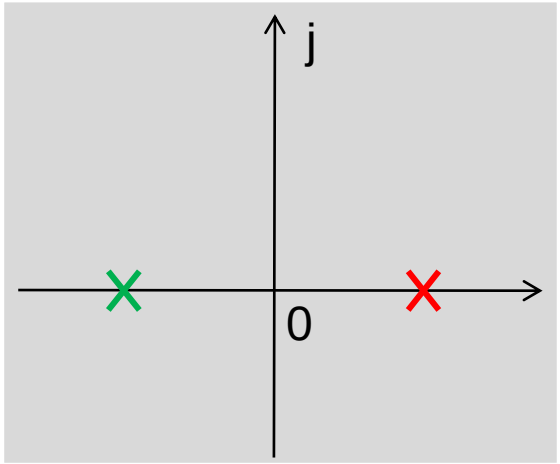
$$\frac{12\varepsilon - 48}{\varepsilon} = 12 - \frac{48}{\varepsilon} < 0$$

The system is unstable and has two poles which are not in the left-half s-plane.

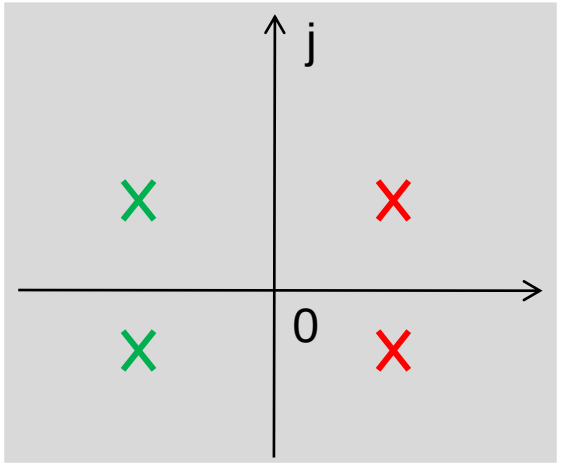
Case 2: an entire row of Routh's tabulation is zero.

This indicates…

There are pairs of roots that are mirror images of each other with respect to the origin.

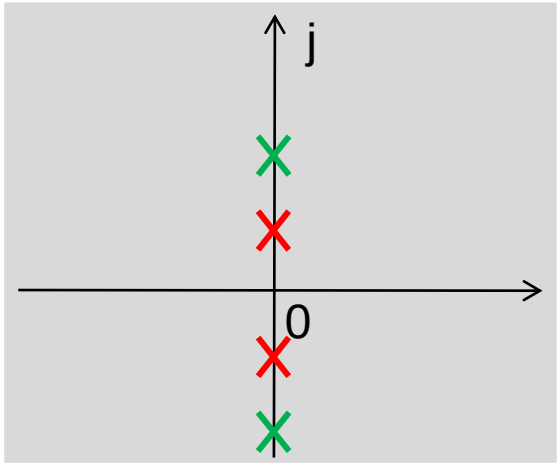


e.g. $s_{1,2} = \pm 1$



e.g. $s_{1,2} = -1 \pm j1$

$s_{3,4} = 1 \pm j1$



e.g. $s_{1,2} = \pm 1j$

$s_{3,4} = \pm 2j$

Example

The characteristic equation of a system is:

$$s^5 + 3s^4 + 3s^3 + 9s^2 - 4s - 12 = 0$$

Determine whether there are any roots on the imaginary axis or in the RHP.

s^5	1	3	−4
s^4	3	9	−12
s^3	12	18	0
s^2	9/2	−12	
s^1	50	0	
s^0	−12		

= 0

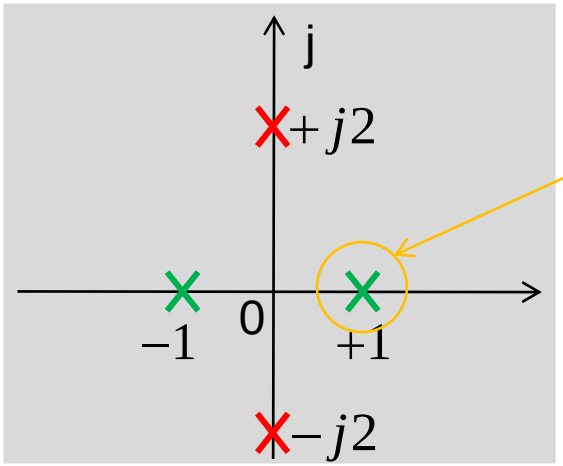
The sign in the first column changes once, so the system is unstable and there is one root outside LHP.

Solving the auxiliary equation

$$A(s) = 3s^4 + 9s^2 - 12 = 0$$

$$s^4 + 3s^2 - 4 = (s^2 - 1)(s^2 + 4) = 0$$

$$s_{1,2} = \pm 1 \qquad s_{3,4} = \pm j2$$



A positive real root locates in the RHP

In-class exercises

Determine the stability of the following systems:

(1) $2s^4 + s^3 + 3s^2 + 5s + 10 = 0$

(2) $s^5 + 3s^4 + 12s^3 + 24s^2 + 32s + 48 = 0$

In-class exercises

Determine the stability of the following system:

$$2s^4 + s^3 + 3s^2 + 5s + 10 = 0$$

s^4	2	3	10
s^3	1	5	
s^2	-7	10	
s^1	$\frac{45}{7}$		
s^0	10		



Two sign changes – two roots in the RHP.

In-class exercises

Determine the stability of the following system:

$$s^5 + 3s^4 + 12s^3 + 24s^2 + 32s + 48 = 0$$

s^5	1	12	32	
s^4	3	24	48	
s^3	4	16		
s^2	12	48		$\rightarrow A(s) = 12s^2 + 48 = 0$
s^1	0			$\leftarrow$ *Entire row zero*
s^1	24			
s^0	48			

Auxiliary polynomial

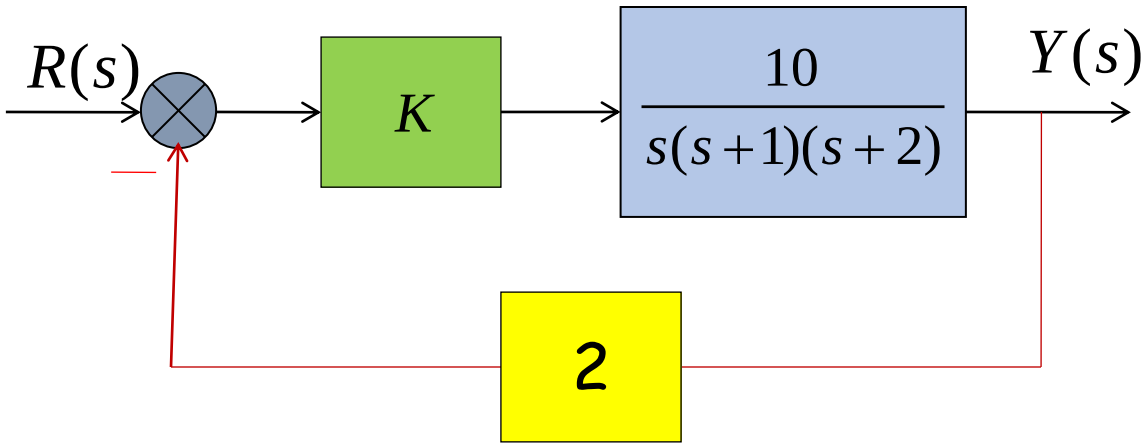
No sign changes – no roots in the RHP. Entire row is zero – there are symmetric roots wrt the origin. The order of the auxiliary polynomial is 2 – two roots are symmetric wrt the origin. Therefore, the system has two (simple) roots on the $j\omega$-axis.

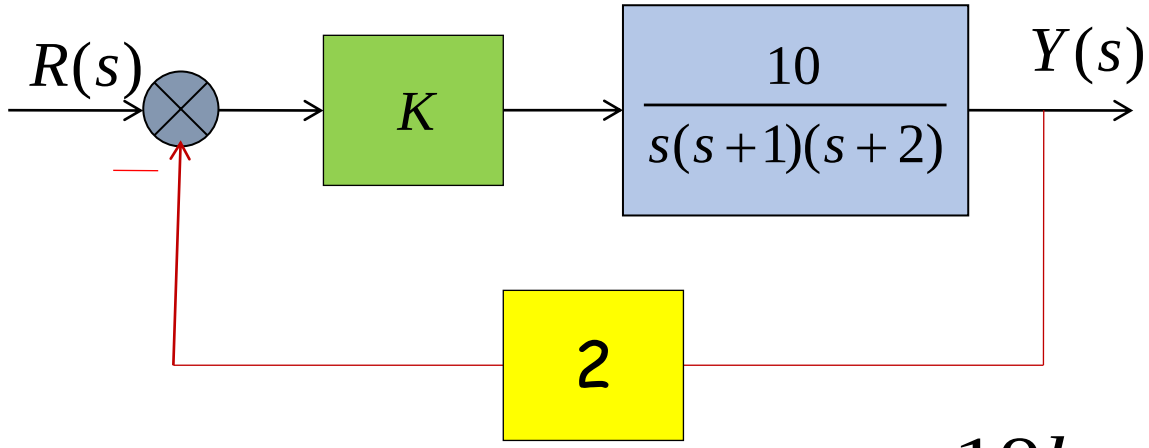
Application of Routh Tabulation

(1) Routh's tabulation can be used not only to determine the stability of a system, but also the spread of its characteristic roots.

(2) For a control system with a regulator, Routh's tabulation can be used to select parameter values so that the system is stable.

Example Determine the range of K so that the system is stable.





$$T(s) = \frac{Y(s)}{R(s)} = \frac{G_{forward}(s)}{1 + G_{loop}(s)} = \frac{\dfrac{10k}{s(s+1)(s+2)}}{1 + \dfrac{20k}{s(s+1)(s+2)}}$$

Characteristic equation:

$$1 + G_{loop}(s) = 0$$

$$\therefore 1 + \frac{20k}{s(s+1)(s+2)} = 0$$

Characteristic equation $1+\frac{20k}{s(s+1)(s+2)}=0$

$$s^3+3s^2+2s+20k=0$$

s^3	1	2
s^2	3	$20k$
s^1	$\frac{3\times 2-20k}{3}$	0
s^0	$20k$	

$\Rightarrow\ 6-20k>0\ \Rightarrow\ k<0.3$

$\Rightarrow\ k>0$

So when $0<k<0.3$, the system is stable.

Exercise

Consider that a 3rd-order system has the characteristic equation $0.025s^3 + 0.325s^2 + s + k = 0$

Determine the range of k so that the system is stable.

Solution. Reformulate the characteristic equation as

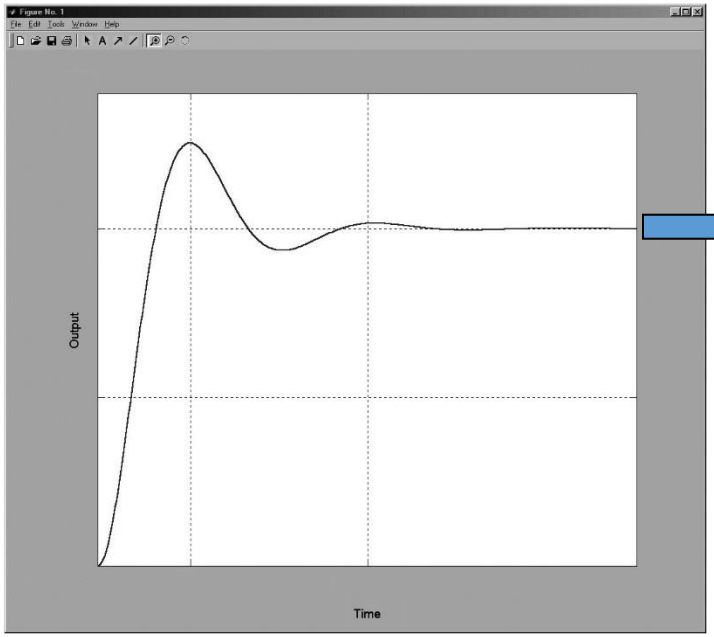
$$s^3 + 13s^2 + 40s + 40k = 0$$

s^3	1	40
s^2	13	$40k$
s^1	$\dfrac{13 \times 40 - 40k}{13}$	$\Rightarrow k < 13$
s^0	$40k$	$\Rightarrow k > 0$

Steady-state error

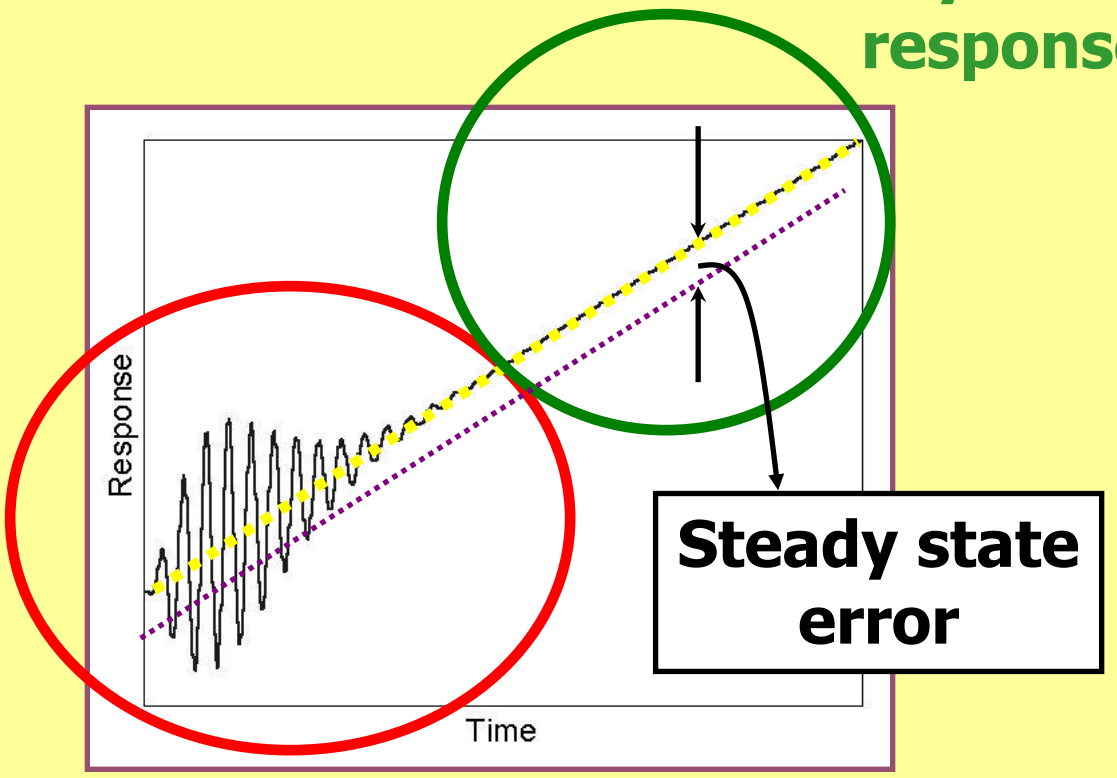
Steady-state error

Steady state response is the manner in which the system output behaves as time approaches infinity.



This is the steady state value

Steady-state error



Steady-state error

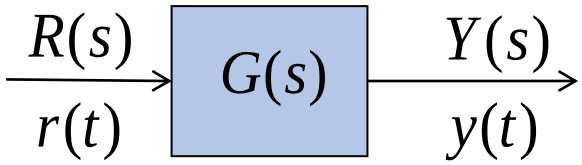
Control systems can be classified according to their ability to follow several test inputs.

We will analyze the steady state error for certain types of inputs, such as step, ramp or parabolic commands.

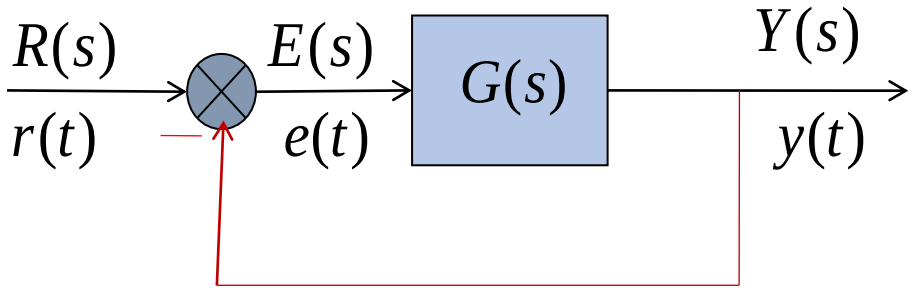
Most input signals can be written as combinations of these signals, so the classification is reasonable.

Error and steady-state error

Open-loop control system



Closed-loop control system



Instantaneous error: $e(t) = r(t) - y(t)$

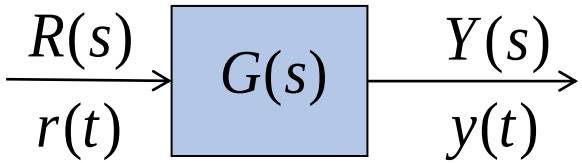
Steady-state error: $e_{ss} = \lim_{t \to \infty} e(t)$

Utilizing the final value theorem: $\lim_{t \to \infty} f(t) = \lim_{s \to 0} sF(s)$

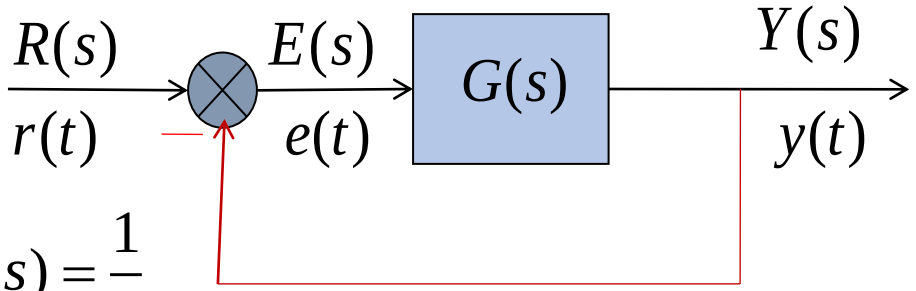
$$e_{ss} = \lim_{t \to \infty} e(t) = \lim_{s \to 0} sE(s)$$

For any given input (such as unit step and unit ramp) one can calculate the steady-state error for given open-loop and closed-loop control systems.

Open-loop control system



Closed-loop control system



Consider unit-step input r(t)=1(t), $R(s)=\frac{1}{s}$

$$E(s)=R(s)-Y(s)$$
$$=R(s)-G(s)R(s)$$
$$=[1-G(s)]R(s)$$
$$e_{ss}=\lim_{s\to 0} sE(s)$$
$$=\lim_{s\to 0} s[1-G(s)]\frac{1}{s}$$
$$=\lim_{s\to 0}[1-G(s)]$$
$$=1-G(0)$$

$$E(s)=R(s)-Y(s)$$
$$=R(s)-\frac{G(s)}{1+G(s)}R(s)$$
$$=\frac{1}{1+G(s)}R(s)$$
$$e_{ss}=\lim_{s\to 0} sE(s)=\lim_{s\to 0} s\frac{1}{1+G(s)}\frac{1}{s}$$
$$=\lim_{s\to 0}\frac{1}{1+G(s)}=\frac{1}{1+G(0)}$$

Feedback has the effect of reducing the steady-state error.

The forward-path (open loop) transfer function G(s) can be formulated as

N is the order of the pole of G(s) at s=0

$$G(s)=\frac{K(\tau_1 s+1)\cdots(\tau_2^2 s^2+2\xi_1\tau_2 s+1)}{s^N(T_1 s+1)\cdots(T_2^2 s^2+2\xi_2 T_2 s+1)}=\frac{K}{s^N}G_0(s)$$

$G_0(s)$ above is

$$G_0(s)=\frac{(\tau_1 s+1)\cdots(\tau_2^2 s^2+2\xi_1\tau_2 s+1)}{(T_1 s+1)\cdots(T_2^2 s^2+2\xi_2 T_2 s+1)},\quad \text{when } s\to 0,\quad G_0(s)\to 1$$

System Type: the order of the pole of G(s) at s=0.

N=0 the system is called type 0,
N=1 the system is called type 1,
N=2 the system is called type 2,

K is called the open-loop gain

Consider the error equation

$$E(s) = \frac{1}{1+G(s)} R(s)$$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s \frac{1}{1+\frac{K}{s^N} G_0(s)} R(s)$$

$$= \lim_{s\to 0} s \frac{1}{1+\frac{K}{s^N}} R(s)$$

Steady state error e_{ss} depends on

- the input (i.e., R(s)) and
- the type of the open loop transfer function (i.e., the value of N)

Consider a unit step input: $r(t) = 1(t), \quad R(s) = \frac{1}{s}$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s \frac{1}{1+\frac{K}{s^N}G_0(s)} \frac{1}{s} = \lim_{s\to 0} \frac{1}{1+\frac{K}{s^N}}$$

N=0, type 0 system $\Rightarrow$ $e_{ss} = \lim_{s\to 0} \frac{1}{1+\frac{K}{s^0}} = \frac{1}{1+K}$

N=1, type 1 system $\Rightarrow$ $e_{ss} = \lim_{s\to 0} \frac{1}{1+\frac{K}{s^1}} = \lim_{s\to 0} \frac{s}{s+K} = 0$

N=2, type 2 system $\Rightarrow$ $e_{ss} = \lim_{s\to 0} \frac{1}{1+\frac{K}{s^2}} = \lim_{s\to 0} \frac{s^2}{s^2+K} = 0$

Consider a unit ramp input: $r(t) = t, \quad R(s) = \frac{1}{s^2}$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s \frac{1}{1+\frac{K}{s^N}G_0(s)} \frac{1}{s^2} = \lim_{s\to 0} \frac{1}{s\left(1+\frac{K}{s^N}\right)}$$

N=0, type 0 system $\Rightarrow$ $e_{ss} = \lim_{s\to 0} \frac{1}{s\left(1+\frac{K}{s^0}\right)} = \infty$

N=1, type 1 system $\Rightarrow$ $e_{ss} = \lim_{s\to 0} \frac{1}{s\left(1+\frac{K}{s^1}\right)} = \lim_{s\to 0} \frac{1}{s+K} = \frac{1}{K}$

N=2, type 2 system $\Rightarrow$ $e_{ss} = \lim_{s\to 0} \frac{1}{s\left(1+\frac{K}{s^2}\right)} = \lim_{s\to 0} \frac{s}{s^2+K} = 0$

Consider a parabolic input: $r(t)=\frac{1}{2}t^2, \quad R(s)=\frac{1}{s^3}$

$$e_{ss}=\lim_{s\to 0} sE(s)=\lim_{s\to 0} s\frac{1}{1+\frac{K}{s^N}G_0(s)}\frac{1}{s^3}=\lim_{s\to 0}\frac{1}{s^2\left(1+\frac{K}{s^N}\right)}$$

N=0, type 0 system ⇒ $e_{ss}=\lim_{s\to 0}\frac{1}{s^2\left(1+\frac{K}{s^0}\right)}=\infty$

N=1, type 1 system ⇒ $e_{ss}=\lim_{s\to 0}\frac{1}{s^2\left(1+\frac{K}{s^1}\right)}=\lim_{s\to 0}\frac{1}{s(s+K)}=\infty$

N=2, type 2 system ⇒ $e_{ss}=\lim_{s\to 0}\frac{1}{s^2\left(1+\frac{K}{s^2}\right)}=\lim_{s\to 0}\frac{1}{s^2+K}=\frac{1}{K}$

N=0, type 0 system

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{1}{1+\frac{K}{s^0}G_0(s)}R(s) = \lim_{s\to 0} s\frac{1}{1+K}R(s)$$

Step input:

$$r(t) = 1(t) \quad R(s) = \frac{1}{s}$$

$$e_{ss} = \frac{1}{1+K}$$

Steady-state error exists and is finite.

Ramp input:

$$r(t) = t \quad R(s) = \frac{1}{s^2}$$

$$e_{ss} = \infty$$

Infinite steady-state error.

Parabolic input:

$$r(t) = \frac{1}{2}t^2 \quad R(s) = \frac{1}{s^3}$$

$$e_{ss} = \infty$$

Infinite steady-state error.

N=1, type 1 system

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{1}{1+\frac{K}{s^1}G_0(s)}R(s) = \lim_{s\to 0} s\frac{1}{1+\frac{K}{s}}R(s)$$

Step input:

$r(t) = 1(t) \quad R(s) = \frac{1}{s}$

$$e_{ss} = \frac{1}{1+\infty} = 0$$

No steady-state error

Ramp input:

$r(t) = t \quad R(s) = \frac{1}{s^2}$

$$e_{ss} = \frac{1}{K}$$

Steady-state error exists

Parabolic input:

$r(t) = \frac{1}{2}t^2 \quad R(s) = \frac{1}{s^3}$

$$e_{ss} = \infty$$

Infinite steady-state error.

Type-1 system can track step signal without error.

N=2, type 2 system

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{1}{1+\frac{K}{s^2}G_0(s)}R(s) = \lim_{s\to 0} s\frac{1}{1+\frac{K}{s^2}}R(s)$$

Step input:

$r(t) = 1(t) \quad R(s) = \frac{1}{s}$

$$e_{ss} = \frac{1}{1+\infty} = 0$$

No steady-state error

Ramp input:

$r(t) = t \quad R(s) = \frac{1}{s^2}$

$$e_{ss} = 0$$

No steady-state error

Parabolic input:

$r(t) = \frac{1}{2}t^2 \quad R(s) = \frac{1}{s^3}$

$$e_{ss} = \frac{1}{K}$$

Steady-state error exists

Type-2 system can track step and ramp signals without error.

Steady-state error constants

Define

$$K_p = \lim_{s \to 0} G(s)$$ - proportional error constant

$$K_v = \lim_{s \to 0} sG(s)$$ - velocity error constant

$$K_a = \lim_{s \to 0} s^2 G(s)$$ - acceleration error constant

Steady-state error constants

N=0, type 0 system

- proportional error constant

$$K_p = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{K}{s^0} G_0(s) = K$$

- velocity error constant

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} s\frac{K}{s^0} G_0(s) = 0$$

- acceleration error constant

$$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} s^2 \frac{K}{s^0} G_0(s) = 0$$

Steady-state error constants

N=1, type 1 system

- proportional error constant

$$K_p = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{K}{s^1} G_0(s) = \infty$$

- velocity error constant

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} s\frac{K}{s^1} G_0(s) = K$$

- acceleration error constant

$$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} s^2 \frac{K}{s^1} G_0(s) = 0$$

Steady-state error constants

N=2, type 2 system

- proportional error constant

$$K_p = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{K}{s^2} G_0(s) = \infty$$

- velocity error constant

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} s\frac{K}{s^2} G_0(s) = \infty$$

- acceleration error constant

$$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} s^2 \frac{K}{s^2} G_0(s) = K$$

Steady-state error in terms of error constants

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{1}{1+G(s)}R(s)$$

Step input: $r(t) = 1(t), \quad R(s) = \frac{1}{s}$

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+G(s)}\frac{1}{s} = \frac{1}{1+\lim_{s\to 0} G(s)} = \frac{1}{1+K_p}$$

Ramp input: $r(t) = t, \quad R(s) = \frac{1}{s^2}$

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+G(s)}\frac{1}{s^2} = \frac{1}{\lim_{s\to 0} sG(s)} = \frac{1}{K_v}$$

Parabolic input: $r(t) = \frac{1}{2}t^2, \quad R(s) = \frac{1}{s^3}$

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+G(s)}\frac{1}{s^3} = \frac{1}{\lim_{s\to 0} s^2G(s)} = \frac{1}{K_a}$$

Steady-state error in terms of error constants

Type of System	Error constants			Steady-state error e_{ss}		
N	K_p	K_v	K_a	$r(t) = R_0 \cdot 1(t)$	$r(t) = V_0 t$	$r(t) = A_0 t^2/2$
0	K_p	0	0	$\frac{R_0}{1+K_p}$	∞	∞
I	∞	K_v	0	0	$\frac{V_0}{K_v}$	∞
II	∞	∞	K_a	0	0	$\frac{A_0}{K_a}$

These hold for unit-feedback systems (H(s)=1)